Urban Parking Behavior and Decision-Making through a Conjoint Analysis Approach

A GIS-Based Spatial Analysis of Parking Imbalance and Accessibility in Dubai

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Submitted in partial fulfillment of the requirements for the degree of
Master of Science in Data Science

January 2026

Dedication

This thesis is dedicated to my daughter, **Shahidi Wangui**, whose presence has been a constant source of strength, inspiration, and motivation throughout my academic journey.

I also dedicate this work to the memory of my late father, **Rocket**. His guidance, values, and belief in education continue to inspire my pursuit of knowledge and perseverance in every challenge.

Abstract

Rapid urbanization and automobile dependence place increasing pressure on urban parking systems, shaping how drivers respond to supply constraints, accessibility conditions, and localized competition for space.

This thesis examines urban parking behavior and decision-making in Dubai through a spatial analytical framework that integrates GIS-based feature engineering, spatial statistics, and supervised machine learning to identify structural patterns of parking imbalance.

Using Dubai as the primary study area, parking capacity was aggregated to a uniform 250 m grid and normalized against urban activity intensity derived from Points-of-Interest density.

Global Moran's I ($I = 0.70$, $p < 0.001$) confirmed strong spatial autocorrelation of imbalance values, while Getis–Ord G_i^* identified statistically significant hotspot regimes concentrated along high-accessibility corridors.

A calibrated Random Forest classifier achieved 94.4% accuracy with improved hotspot precision (0.67), demonstrating that structural accessibility variables contain strong predictive signal.

A cross-city transfer experiment revealed limited generalizability, highlighting spatial non-stationarity in parking determinants.

The framework is further interpreted within a broader urban parking decision-support perspective, in which empirically validated hotspot detection can support policy-aligned reasoning and future AI-enabled governance extensions.

The study contributes a scalable analytical framework for examining urban parking behavior and decision-making through spatial inference, predictive analytics, and decision-support interpretation in data-constrained urban environments.

Keywords: Spatial Autocorrelation, Parking Imbalance, Urban Accessibility, Machine Learning, Random Forest, Decision Support, AI Governance, Dubai Mobility

Declaration

I hereby declare that this thesis is my own work and that all the sources used have been acknowledged. The use of artificial intelligence tools has been declared separately in accordance with university regulations.

Acknowledgments

I would like to express my deepest gratitude to my main supervisor, Dr. Rafael Milani Medeiros, for his invaluable guidance, continuous encouragement, and academic mentorship throughout the development of this thesis. His thoughtful feedback, methodological insight, and support at critical stages of the research were instrumental in shaping the direction and quality of this work. I am sincerely grateful for his patience and dedication in supervising this study.

I would also like to extend my sincere appreciation to my second supervisor, Prof. Dahlia Mahmoud, for her support, constructive comments, and academic perspective during the course of this research. Her contributions and encouragement have been greatly appreciated.

I am also thankful to the faculty of the Master's program for providing the academic environment and methodological foundations that made this research possible.

Finally, I would like to express my heartfelt gratitude to my family for their constant support, encouragement, and understanding throughout my academic journey. Their belief in my efforts has been a continuous source of motivation.

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Chapter 1

Introduction

1.1 Background

Dubai has experienced one of the fastest urban expansions in the Middle East over the past three decades. The emirate's population has grown from approximately 689,000 in 1995 to over 3.6 million residents by 2023, reflecting sustained economic diversification, infrastructure investment, and global migration inflows (Dubai Statistics Center, 2023).

Motorization levels have increased in parallel with demographic and economic growth. Estimates indicate that Dubai maintains more than 540 vehicles per 1,000 residents, placing it among the highest motorization rates in the region (Roads and Transport Authority (RTA), 2023c; World Bank, 2023). Daily vehicle movements exceed 3.5 million trips, exerting sustained pressure on road capacity and parking infrastructure (Roads and Transport Authority (RTA), 2023a).

Although the Roads and Transport Authority (RTA) has expanded metro coverage, intelligent transport systems, and multimodal integration strategies(Roads and Transport Authority (RTA), 2021; Dubai Government, 2022), private vehicles continue to account for a substantial share of commuter and commercial mobility. As a result, parking infrastructure functions as a critical intermediary layer within the broader urban accessibility system and influences how urban users respond to localized constraints of access, proximity, and availability.

From a theoretical perspective, accessibility-driven land-use intensity is a primary determinant of spatial economic concentration (Hansen, 1959). Urban activity clusters form around nodes of commercial density, retail agglomeration, and mixed-use vertical development. Consistent with spatial interaction theory(Wilson, 1967) and Tobler's

First Law of Geography(Tobler, 1970), spatial demand patterns exhibit structured autocorrelation rather than random dispersion.

Dubai's parking infrastructure consists of regulated on-street zones and structured off-street facilities. Despite expansion of capacity, spatial mismatch between activity intensity and supply allocation persists, particularly within high-density commercial corridors. Such imbalance reflects not only supply constraints but also the spatial concentration of economic functions (Zakharenko, 2016; Gragera and Albalate, 2021).

Environmental externalities reinforce the urgency of systematic parking analysis. Cruising behavior increases congestion, fuel consumption, and localized emissions, as documented in empirical studies on parking search dynamics (Shoup, 2006; Tang et al., 2014). Inefficient parking allocation therefore produces measurable mobility and environmental spillovers.

Urban parking behavior and decision-making can be examined through multiple analytical lenses, including direct preference-based approaches and observation of real-world spatial conditions. In practice, parking-related decisions are shaped by the interaction between supply, accessibility, surrounding land-use intensity, and localized competition for space. These dynamics become visible not only through stated choices, but also through recurring spatial patterns that reveal where pressure, imbalance, and constrained access are most pronounced.

In response to broader mobility challenges, Dubai has initiated AI-driven transport strategies focused on predictive congestion modeling and smart mobility governance (Roads and Transport Authority (RTA), 2023b). However, fine-grained analysis of parking-related spatial pressure remains comparatively underdeveloped, particularly at standardized neighbourhood resolution and in forms that can support the interpretation of parking behavior and urban decision environments.

This context motivates the development of a grid-based analytical framework for examining urban parking behavior and decision-making through observed spatial patterns. The framework is designed to:

- identify statistically significant parking imbalance regimes,
- classify and predict hotspot zones using engineered spatial features,
- integrate multi-source geospatial evidence to strengthen spatial interpretation, and
- support policy-relevant understanding of parking-related decision environments.

1.2 Problem Statement

Urban parking infrastructure plays a structurally significant yet analytically underdeveloped role within metropolitan mobility systems. While road congestion, traffic flow forecasting, and public transport optimization have received substantial attention within intelligent transport research, parking systems remain comparatively under-examined despite their direct influence on accessibility, congestion spillovers, emissions, and land-use allocation.

In rapidly expanding cities such as Dubai, high levels of motorization,concentrated commercial development, and mixed-use vertical growth generate localized parking demand peaks. Although structured parking capacity exists within many districts, spatial mismatch between supply and activity intensity produces concentrated imbalance zones. These zones are not randomly distributed but are shaped by accessibility gradients and spatial dependence structures, as demonstrated in spatial autocorrelation theory(Getis and Ord, 1992; Anselin, 1995).

This creates not only an infrastructure problem but also a decision-making problem. Where parking supply is spatially misaligned with surrounding activity intensity, urban users are more likely to encounter constrained choice environments characterized by search costs, access trade-offs, and localized competition for space. Understanding these environments requires analytical approaches capable of linking parking pressure to the broader spatial conditions in which parking-related decisions occur.

Traditional parking analysis methods rely on manual surveys, static administrative inventories, or isolated sensor deployments. While accurate at micro-scale, these approaches are difficult to scale across entire metropolitan regions.They rarely incorporate inferential spatial statistics or predictive modeling frameworks capable of identifying statistically significant imbalance regimes across large urban systems.

Moreover, emerging AI-driven transport initiatives primarily focus on macro-scale congestion modeling rather than parking-related spatial pressure at fine geographic resolution. This creates a methodological gap: limited integration exists between spatial autocorrelation inference, machine learning classification, and urban decision-oriented interpretation within a unified analytical framework.

A further challenge concerns spatial non-stationarity. Models trained in one urban morphology may not generalize across cities due to differences in zoning, mobility patterns, and economic structure. This raises important questions regarding transferability and domain sensitivity in urban analytical systems, particularly where parking behavior and decision-making are shaped by context-specific urban form.

1.3 Research Aim

The aim of this thesis is to examine urban parking behavior and decision-making in Dubai by analyzing how parking supply, accessibility, and land-use intensity interact across space through a GIS-based analytical framework supported by spatial statistics and machine learning.

1.4 Research Objectives

The study pursues the following objectives:

1. To analyze how parking supply and urban accessibility conditions shape spatial patterns associated with parking behavior and parking-related decision-making.
2. To construct a standardized 250 m grid-based GIS framework for aggregating parking supply and activity intensity indicators across Dubai.
3. To quantify structural parking imbalance using a normalized supply–activity index derived from parking capacity and POI-based accessibility proxies.
4. To identify statistically significant spatial clustering patterns that indicate areas of concentrated parking pressure and constrained parking-related decision environments.
5. To evaluate whether engineered spatial variables can be used to classify and predict hotspot regimes associated with intensified parking competition.
6. To assess the extent to which the resulting analytical framework may support evidence-based parking and mobility decision-making in other urban contexts.

1.5 Research Questions

The study addresses the following research questions:

1. How do parking supply, accessibility, and land-use intensity interact to shape spatial patterns associated with urban parking behavior and decision-making in Dubai?
2. Which spatial factors are most strongly associated with statistically significant parking imbalance clusters?
3. To what extent can GIS-derived and accessibility-related indicators be used to identify and interpret areas of intensified parking pressure?
4. To what extent can this analytical framework support transferable parking-related decision-making in other urban contexts?

1.6 Scope and Significance of the Study

This research examines spatial patterns of formal parking capacity using publicly available geospatial datasets and remotely derived land-use indicators. The analysis is conducted at a fixed 250 m grid resolution to ensure spatial consistency and comparability across districts.

The study does not incorporate transaction-level parking records, sensor-based occupancy streams, or dynamic pricing datasets. Instead, it evaluates structural supply–activity relationships derived from spatial infrastructure and urban intensity proxies.

By avoiding dependence on proprietary sensor systems, the framework emphasizes reproducibility and applicability in contexts where real-time monitoring infrastructure is incomplete, unevenly distributed, or financially constrained.

The contribution lies in demonstrating how statistically validated clustering methods and ensemble classification techniques grounded in Random Forest theory (Breiman, 2001) can be combined within a spatially standardized analytical framework to generate interpretable parking imbalance diagnostics relevant to urban parking behavior and decision-making.

Random Forests provide robustness to nonlinear feature interactions and multicollinearity, making them particularly suitable for heterogeneous urban feature spaces.

This approach provides a replicable analytical blueprint for municipalities seeking to improve parking governance through data-informed spatial prioritization and decision support rather than purely reactive enforcement strategies.

1.7 Structure of the Dissertation

The dissertation is organized into five chapters.

Chapter 1 introduces the research problem, contextual background, objectives, and overall contribution.

Chapter 2 reviews relevant literature on parking systems, spatial autocorrelation methods, machine learning applications in mobility analysis, urban accessibility theory, and parking-related decision environments in urban contexts.

Chapter 3 details the methodological design, including spatial unit construction, feature engineering, statistical clustering procedures, predictive model development, and the analytical workflow used to examine parking imbalance patterns.

Chapter 4 presents empirical findings, model evaluation metrics, cross-domain testing outcomes, and spatial interpretation of imbalance regimes.

Chapter 5 discusses theoretical implications, policy considerations, limitations, and directions for future research.

Chapter 2

Literature Review

Thematic Structure and Analytical Positioning of the Literature

The literature reviewed in this study spans multiple disciplinary domains, including spatial statistics, urban economics, remote sensing, transportation engineering, and machine learning. Rather than presenting an annotated bibliography, the review was organized through a thematic synthesis approach. This approach was adopted to identify methodological patterns, theoretical assumptions, and analytical gaps across studies, and to structure the review around conceptual building blocks directly relevant to the proposed framework.

Following an iterative reading process, five dominant methodological themes emerged:

1. Spatial Statistical Approaches to Urban Parking and Mobility Hotspot Analysis
2. GIS-Based Accessibility and Feature Engineering
3. Remote Sensing and Satellite–GIS Hybrid Modeling
4. Machine Learning for Parking and Urban Intensity Prediction
5. Cross-City Transferability and Spatial Non-Stationarity

These themes were not predefined. Instead, they were derived inductively through systematic comparison of model architectures, statistical methods, data sources, and evaluation strategies across the reviewed studies. Papers were grouped according to their primary analytical contribution rather than their application domain alone.

The first theme captures foundational spatial statistics used to identify clustered urban phenomena. The second focuses on structural determinants of parking intensity derived from accessibility and land-use variables. The third addresses advances in satellite-derived feature extraction and raster–vector integration. The fourth synthesizes predictive modeling approaches, including ensemble and deep learning architectures. The fifth examines domain adaptation and cross-city transferability, an area of growing importance in data-scarce urban contexts.

This thematic structuring reflects the analytical logic of the present thesis. The proposed methodology integrates

- (Theme 1) Spatial autocorrelation analysis,
- (Theme 2) Accessibility-based feature engineering,
- (Theme 3) Satellite-derived land-use covariates,
- (Theme 4) Supervised machine learning,
- (Theme 5) Empirical evaluation of cross-city transferability.

By organizing the literature around methodological convergence rather than disciplinary boundaries, this chapter establishes a coherent foundation for the integrated spatial-statistical modeling framework developed in Chapter 3.

Literature Search Strategy and Selection Criteria

The literature review was conducted through a structured search strategy across major academic databases, including Scopus, Web of Science, IEEE Xplore, ScienceDirect, and Google Scholar. Additional cross-referencing was performed using Semantic Scholar to identify highly cited and methodologically influential studies.

The literature review evolved iteratively alongside the methodological development of the study. Initial searches focused broadly on “urban parking modeling” and “parking occupancy prediction”. As the analytical framework developed, subsequent searches expanded to include terms related to spatial clustering, accessibility theory, remote sensing integration, and machine learning classification.

Additional targeted searches were conducted after identifying relevant statistical techniques within influential papers. For example, once spatial autocorrelation methods emerged as methodologically relevant, focused searches were performed on "Getis–Ord G_i^*", "Local Moran's I", and "spatial hotspot analysis". Similarly, when satellite-derived land-use integration became part of the feature engineering strategy, searches were extended to include "Sentinel-2 urban analysis" and "land-cover classification".

Boolean operators were used selectively to refine results (e.g., "parking AND spatial statistics", "urban accessibility AND machine learning"). This iterative process ensured that the literature review remained aligned with the evolving analytical design rather than being constrained by predefined thematic categories.

Inclusion criteria required that studies:

- Present explicit methodological frameworks (statistical or machine learning).
- Provide sufficient technical detail to allow replication or adaptation.
- Include quantitative evaluation metrics.
- Be published in peer-reviewed journals or reputable conference proceedings.

Studies focusing exclusively on conceptual discussions without empirical implementation were excluded unless they provided foundational theoretical models relevant to parking economics or spatial dependence.

Preference was given to publications from 2015 onward to ensure methodological relevance, while seminal works(Getis and Ord, 1992; Anselin, 1995) were retained due to their foundational importance in spatial statistics.

This structured search and filtering process ensured that the reviewed literature was both methodologically rigorous and directly aligned with the analytical objectives of this thesis.

This adaptive search strategy reflects the interdisciplinary nature of urban parking research, where statistical, economic, remote sensing, and artificial intelligence literatures intersect.

2.1 Spatial Statistical Approaches to Urban Parking and Mobility Hotspot Analysis

Understanding spatial concentration patterns is central to urban parking analysis. Urban infrastructure systems are rarely randomly distributed; instead, they exhibit structured clustering driven by accessibility, land use, and economic intensity. Spatial autocorrelation statistics provide a rigorous inferential framework for detecting statistically significant clustering beyond descriptive heatmaps.

Among the most widely adopted inferential statistics are the Getis–Ord G_i^* statistic and Local Moran's I, both of which enable localized identification of high-intensity (hotspot) and low-intensity (coldspot) clusters.

Table 2.1: Key Studies on Spatial Hotspot Detection Methods

Authors	**Year**	**Methods**	**Data Used**	**Geography**
Getis, A.; Ord, J.K.	1992	Global G, Local G_i^*	Spatial point data	USA
Anselin, L.	1995	Local Moran's I (LISA)	Areal data	Conceptual
Hovenden, E.	2020	Moran's I, G_i^*, KDE	Road crash data	Melbourne, Australia
Crimmins, M. et al.	2021	G_i^*, Local Moran's I	Near-collision events	USA

Getis and Ord (1992) introduced the G_i^* statistic for identifying statistically significant spatial clustering using distance-based weights. The statistic produces standardized z-scores, enabling classification of hot and cold spots. However, applications remain largely descriptive.

Anselin (1995) developed Local Indicators of Spatial Association (LISA), including Local Moran's I, enabling localized cluster detection and spatial outlier identification. Despite its methodological strength, applications are often not integrated into predictive modeling frameworks.

Hovenden et al. (2020) demonstrated how different spatial weight specifications influence hotspot outcomes in road crash data. Crimmins et al. (2021) similarly applied G_i^* to mobility safety analysis but did not extend results into forecasting systems.

Synthesis of Theme The literature establishes spatial autocorrelation statistics as statistically rigorous tools for identifying structured clustering. However, few studies operationalize hotspot outputs as dependent variables within supervised machine learning systems.

Applications to parking infrastructure, particularly in rapidly urbanizing regions, remain underexplored.

Research Gap Existing spatial hotspot literature lacks integration with predictive modeling and cross-city generalization, particularly for parking systems in Middle Eastern and Sub-Saharan contexts.

2.2 GIS-Based Accessibility and Urban Feature Engineering

Parking intensity is shaped by structural determinants such as accessibility, connectivity, and economic activity concentration. Contemporary GIS-based research emphasizes feature engineering approaches that translate these determinants into measurable spatial indicators.

Table 2.2: Key Studies on Accessibility and Parking Demand Modeling

Authors	Year	Methods	Data Used	Geography
Shoup, D.	2006	Cruising cost theory	Parking observations	USA
Zakharenko, R.	2016	Congestion-pricing model	Urban parking data	USA
Gragera, A.; Albalate, D.	2021	Spatial econometric model	On-street parking	Spain
Xiong, S. et al.	2025	Sentinel-2 + POI fusion	Satellite + POIs	Global cities

Shoup (2006) formalized the concept of cruising externalities arising from systematically underpriced curb parking. The core argument states that when on-street parking prices are set below market-clearing levels, drivers engage in circulatory search behavior rather than utilizing available off-street alternatives.

This behavior generates negative externalities including congestion, increased fuel consumption, localized emissions, and time inefficiencies. Importantly, the study reframes parking not merely as a land-use allocation issue but as a dynamic urban traffic phenomenon with measurable welfare impacts.

The cruising model demonstrates that parking mispricing creates spatial concentration of search activity in high-demand corridors, effectively producing localized pressure zones that resemble statistically detectable hotspots.

Zakharenko (2016) extended this theoretical framing by embedding parking pricing within equilibrium congestion models. Rather than treating parking search as an isolated behavioral anomaly, the equilibrium framework links pricing, spatial substitution, and congestion dynamics across competing parking facilities.

Under this formulation, parking demand redistributes spatially in response to relative price gradients, producing substitution effects between curbside and off-street supply. This equilibrium perspective provides a formal explanation for spatial imbalance patterns, as underpriced zones attract excess demand that propagates spillover effects into adjacent streets.

Building upon this, Gragera and Albalate (2021) introduced spatial econometric modeling to explicitly capture spillover dependencies between neighboring parking segments. Using spatial lag and error specifications, the study demonstrated that parking occupancy in one street segment is statistically influenced by adjacent segments, confirming the presence of endogenous spatial feedback mechanisms.

This finding is critical because it empirically validates the assumption that parking systems exhibit spatial dependence rather than independent allocation.The incorporation of spatial weights matrices provides a methodological bridge between economic parking theory and formal spatial statistics.

More recently,(Xiong et al., 2025) demonstrated the integration of satellite-derived land-use indicators with points-of-interest (POI) data to construct hybrid GIS–remote sensing feature spaces. By fusing raster-based built-environment characteristics with vector-based activity intensity proxies, the study illustrated how urban economic concentration can be operationalized at fine spatial resolutions.

This methodological fusion is particularly relevant in data-sparse environments where sensor-based occupancy measurements are unavailable. The integration of satellite-derived morphology with POI density provides a potentially transferable basis for estimating structural demand intensity and supporting spatial classification models.

Synthesis of Theme Accessibility and land-use composition are core determinants of parking intensity. However, integration between econometric demand models, satellite-derived covariates, and supervised spatial classification remains limited.

Research Gap Relatively few studies identified in this review combine accessibility metrics, land-use proportions, and hotspot-derived labels within unified machine learning frameworks.

2.3 Remote Sensing and Satellite–GIS Hybrid Models

Recent advances in Sentinel-2 imagery and machine-learning-based land-cover classification enable extraction of fine-scale urban morphology indicators.

Beyond static land-cover classification, recent research emphasizes multi-temporal analysis and raster-to-vector aggregation strategies that enable translation of pixel-level spectral information into standardized areal units.

Data cube architectures allow temporal stacking of Sentinel-2 imagery, facilitating extraction of built-up intensity, vegetation fraction, impervious surface proxies, and morphological heterogeneity indices.

These indicators are particularly relevant for parking demand modeling because built-environment density and land-use mix are structural drivers of localized vehicle accumulation. However, most satellite-based studies stop at descriptive land-use mapping and do not extend these features into supervised classification frameworks targeting infrastructure imbalance.

Table 2.3: Key Studies on Satellite–GIS Urban Modeling

Authors	Year	Methods	Data Used	Geography
Wu, S. et al.	2024	Spatio-temporal data cube	Sentinel-2	Global cities
Dynamic World Team	2023	Probabilistic classification	Sentinel-2	Global
Aghazadeh, F.	2025	Raster Gi* analysis	Land surface temp.	Regional
Xiong, S. et al.	2025	Satellite–POI fusion	Sentinel-2 + POIs	Global cities

Wu et al. (2024)constructed a global green space cube, illustrating raster-to-zone aggregation techniques. (Dynamic World Team, 2023) provides probabilistic land-cover classification suitable for grid-level aggregation.

Despite their utility, these datasets are rarely integrated into parking intensity prediction models.

Research Gap Satellite-derived land-use signatures appear underutilized in parking hotspot classification and cross-city predictive modeling within the literature reviewed here.

2.4 Machine Learning for Parking and Urban Intensity Prediction

Machine learning approaches, including ensemble and deep neural architectures, have improved occupancy forecasting performance in many applied studies.

At the foundational level, inferential spatial statistics such as Getis–Ord G_i^* and Local Moran's I (Getis and Ord, 1992; Anselin, 1995) establish statistically significant clustering regimes.These methods ensure that spatial dependence is formally tested rather than implicitly assumed.

Building upon this inferential layer, ensemble learning through Random Forest classification provides predictive generalization across heterogeneous feature spaces. Unlike purely deep neural architectures, ensemble trees offer interpretability through feature importance ranking, supporting transparency in urban planning contexts.

Table 2.4: Key Studies on Machine Learning for Parking Prediction

Authors	**Year**	**Methods**	**Data Used**	**Geography**
Yang, S.; Ma, W.; Pi, X.; Qian, S.	2020	GCNN + LSTM	Sensor networks	China
Ma, C. et al.	2025	EMD + LSTM	Occupancy series	China
Sensors CALM Study	2023	CNN + LSTM clustering	Parking lots	Europe
Abbas, A. et al.	2025	Faster R-CNN optimization	Aerial imagery	Middle East

Xiao et al. (2021) proposed graph convolutional networks integrated with LSTM to capture spatial and temporal dependencies.

Ma et al. (2023) improved prediction accuracy through signal decomposition techniques. However, most studies rely on dense sensor networks and do not incorporate spatial autocorrelation outputs as predictive targets.

Expanded Synthesis The thesis positions Random Forest not as a purely predictive tool, but as a regime classifier grounded in spatial inference.

Research Gap Integration between spatial statistical classification and supervised machine learning appears limited, particularly in cross-city contexts.

2.5 Cross-City Transferability and Spatial Non-Stationarity

Urban systems exhibit spatial non-stationarity, meaning relationships between predictors and outcomes vary geographically.

Spatial non-stationarity implies that relationships between explanatory variables and observed outcomes vary across geographic contexts.

In transportation research, this phenomenon has been documented in traffic flow prediction, crash severity modeling, and demand forecasting,where model coefficients differ significantly between metropolitan regions.

Such variability may arise from differences in urban morphology, economic structure, pricing regimes, or behavioral norms. Failure to account for spatial heterogeneity can lead to model performance degradation when transferred to new domains. Consequently, cross-city validation is not merely a robustness check,but a structural test of model generalization capacity.

Table 2.5: Key Studies on Cross-City Transfer and Domain Adaptation

Authors	**Year**	**Methods**	**Data Used**	**Geography**
Zhang, X. et al.	2025	Domain adaptation	Traffic flow	China
IJCAI STAN Framework	2022	Adversarial transfer	Urban flow	Multi-city
Ionita et al.	2023	Parking clustering	Occupancy data	Europe
Gragera, A.; Albalate, D.	2021	Spatial spillover modeling	Parking data	Spain

Cross-city transfer research remains focused primarily on traffic flow rather than parking systems.

Research Gap Empirical evaluation of cross-city transferability for parking hotspot classification appears limited in the literature reviewed here.

2.6 Computer Vision and Object Detection in Urban Analytics

Deep learning architectures have significantly advanced real-time object detection.

YOLO (You Only Look Once), introduced by Redmon et al. (2016), unified object localization and classification within a single convolutional neural network framework, enabling real-time detection performance. Subsequent iterations (Redmon and Farhadi, 2018) improved bounding-box precision and multi-scale detection.In urban mobility contexts, YOLO architectures have been applied to aerial imagery for vehicle detection and parking occupancy estimation.

Integrating YOLO-based detection with satellite imagery would allow direct vehicle counting, enabling dynamic occupancy validation of statistically inferred hotspot zones.

While not implemented in the present study, this extension represents a key future direction in transitioning from structural imbalance modeling toward real-time parking intelligence systems.

2.7 Retrieval-Augmented AI and Urban Governance

Recent advances in large language models have enabled retrieval-augmented reasoning systems.

Lewis et al. (2020) introduced Retrieval-Augmented Generation (RAG), combining dense document retrieval with generative language models for knowledge-intensive tasks.

Unlike purely generative systems, RAG architectures ground outputs in retrieved documents, improving factual consistency.Transformer architectures underpinning modern language models were introduced by Vaswani et al. (2017), enabling attention-based sequence modeling without recurrence.

In governance contexts, such architectures enable contextual policy interpretation layered atop quantitative analytics.

The conceptual RAG layer proposed in this thesis is intended to extend parking imbalance detection toward policy-aligned decision support by retrieving relevant planning documentation (e.g., RTA strategies (Roads and Transport Authority (RTA), 2021)) and synthesizing structured recommendations. This positions the framework within an emerging paradigm of AI-augmented urban governance systems.

2.8 Consolidated Research Gaps and Conceptual Positioning

The expanded thematic synthesis of the literature reveals a set of methodological, theoretical, and governance-related gaps that collectively motivate the integrated framework developed in this thesis.

Gap 1: Spatial Inference and Predictive Modeling Remain Disconnected

Spatial autocorrelation techniques such as Moran's I and Getis–Ord G_i^* (Getis and Ord, 1992; Anselin, 1995) provide statistically rigorous tools for identifying clustered urban phenomena. However, in most applications,these statistics function as descriptive endpoints rather than as regime generators for supervised learning.

Existing studies rarely operationalize statistically inferred hotspot regimes as dependent variables within predictive machine learning architectures. Consequently, a gap persists between inferential spatial statistics and supervised classification systems.

The present thesis addresses this gap by using G_i^*-derived hotspot labels as the training target for a calibrated Random Forest model, thereby bridging spatial inference and predictive analytics.

Gap 2: Parking Economics and Structural Modeling Remain Separate

The parking literature emphasizes behavioral dynamics such as cruising, search behavior, and congestion externalities (Shoup, 2006; Tang et al., 2014). These works highlight the endogenous feedback loops created by underpriced or misallocated parking supply.

However, empirical integration between parking economics and scalable,grid-based spatial modeling remains limited. Accessibility theory (Hansen, 1959) explains opportunity concentration, yet few studies operationalize this concept within machine learning frameworks using standardized spatial units.

This thesis contributes by translating economic and behavioral theory into a measurable imbalance index constructed from supply–demand differentials at uniform 250 m grid resolution.

Gap 3: Deep Learning and Structural Parking Modeling Are Rarely Integrated

Computer vision systems such as YOLO (Redmon et al., 2016) have demonstrated strong performance in object detection tasks, including vehicle recognition in aerial imagery.

Yet, deep learning-based vehicle detection is typically applied in real-time occupancy monitoring contexts and remains largely disconnected from inferential spatial statistics and structural imbalance modeling.

There is limited integration between pixel-level object detection and zone-based spatial clustering frameworks. Bridging YOLO-based detection with grid-level imbalance inference would allow validation of statistically detected hotspot regimes using dynamic occupancy signals.

Although not implemented directly in this study, this extension represents a clear methodological frontier.

Gap 4: AI Reasoning Systems Are Underutilized in Urban Governance

Recent advances in Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) demonstrate how large language models can be grounded in retrieved knowledge sources to produce contextually aligned outputs.

However, applications of RAG architectures in urban planning and mobility governance remain nascent. Predictive models frequently generate numerical outputs without integrated mechanisms for policy contextualization.

There exists a gap between quantitative spatial analytics and regulatory interpretation.

This thesis proposes a conceptual RAG layer that retrieves relevant planning documentation (e.g., RTA mobility strategies (Roads and Transport Authority (RTA), 2021)) and synthesizes policy-aligned explanations without altering the predictive core.

Gap 5: Urban AI Systems Lack Governance-Layer Structuring

While machine learning and deep learning systems are increasingly deployed in urban contexts, explicit architectural separation between:

- Sensing (data acquisition),
- Statistical inference,

- Predictive modeling,
- Perception (object detection),
- Reasoning (knowledge retrieval and synthesis),

is rarely formalized.Most studies focus on a single analytical layer rather than designing multi-layered AI governance stacks.

The framework proposed in this thesis conceptualizes an integrated urban AI architecture linking spatial statistics, ensemble learning,deep learning extensions, and RAG-based reasoning under explicit governance controls.

Gap 6: Spatial Non-Stationarity and Transferability Remain Underexplored

Urban systems exhibit spatial heterogeneity. Models trained in one morphological context often fail to generalize to another due to distributional shifts.

These distributional shifts reflect both feature-space divergence and regime reclassification under differing urban intensity thresholds.

Although domain adaptation and graph-based learning frameworks (Kipf and Welling, 2017) offer promising directions, empirical evaluation of cross-city transferability in parking imbalance modeling remains limited.

The Seattle application conducted in this thesis demonstrates that structural parking determinants are context-sensitive, reinforcing the importance of localized calibration in urban AI systems.

Conceptual Positioning of the Present Study

In response to these gaps, this thesis positions itself at the intersection of:

- Inferential spatial statistics,
- Accessibility-based demand modeling,
- Supervised ensemble learning,
- Deep learning object detection extensions,

- Retrieval-Augmented policy reasoning,
- AI governance stack architecture.

Rather than treating parking imbalance as either a purely economic,statistical, or machine learning problem, the study proposes a unified analytical stack that integrates detection, prediction, validation, and interpretation.

This positioning extends existing literature by bridging methodological silos and introducing a structured framework for AI-augmented urban mobility governance.

Chapter 3

Methodology

3.1 Study Area

The primary study area is the Emirate of Dubai, United Arab Emirates, a highly urbanized metropolitan region characterized by intensive vehicular mobility, strong economic clustering, and heterogeneous land-use distribution. Dubai was selected due to the availability of structured parking infrastructure datasets, administrative boundary layers, and open-access geospatial resources suitable for scalable spatial modeling.

To evaluate the transferability of the developed modeling framework, a secondary test domain was defined in Seattle, United States. This external application serves as a cross-context validation environment. Seattle data were not used during model training and were preserved exclusively for out-of-sample generalization assessment.

All Dubai-based spatial analyses were conducted in WGS 84 / UTM Zone 40N (EPSG:32640), a projected coordinate reference system appropriate for the longitudinal zone covering the United Arab Emirates. This projection ensures metric consistency for distance-based operations, area calculations, and spatial discretization.

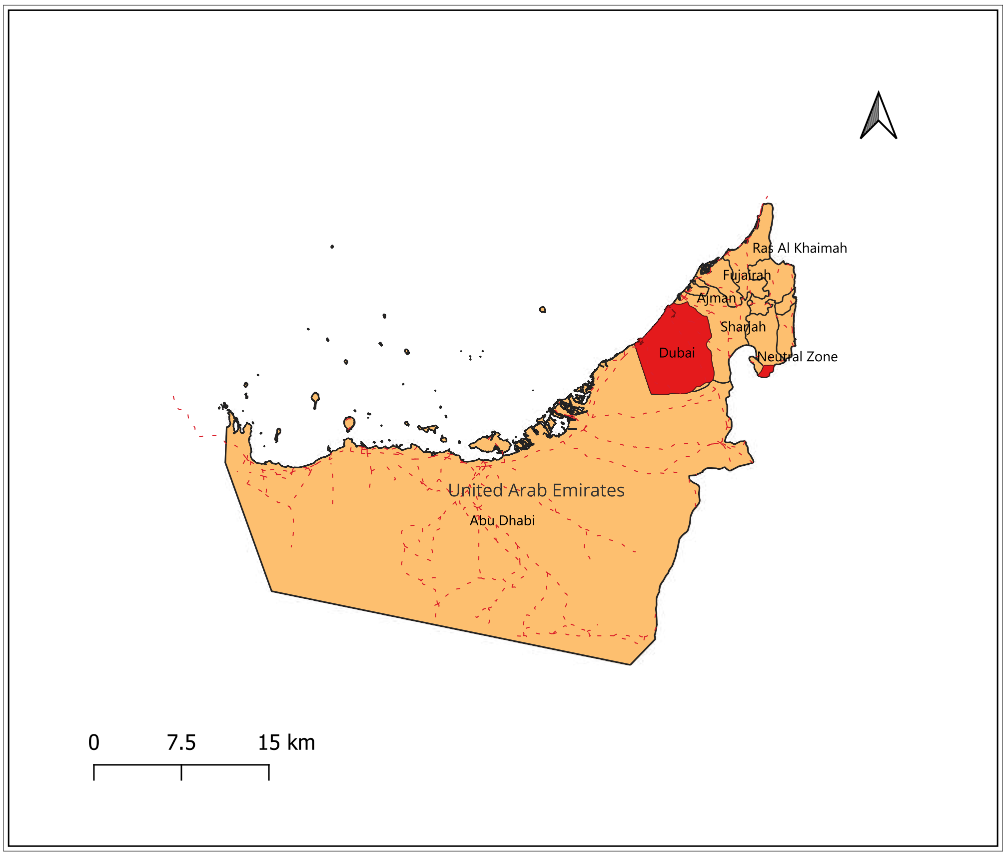


Figure 3.1: Study area showing the Emirate of Dubai within the United Arab Emirates.

For the Seattle application, datasets were reprojected into WGS 84 / UTM Zone 10N (EPSG:32610) to preserve geometric fidelity within the Pacific Northwest UTM zone. Each city was therefore analyzed in its geographically appropriate projected coordinate system to minimize spatial distortion. Spatial Preprocessing and boundary verification were performed using QGIS (Long-Term Release), with selected tasks automated via PyQGIS scripting to ensure reproducibility.

This grid-based discretization reduces bias introduced by irregular administrative units and mitigates the Modifiable Areal Unit Problem (MAUP).

Google Earth historical imagery was additionally used to visually validate temporal urban expansion patterns and parking surface persistence across selected high-intensity zones. These images serve as qualitative temporal validation rather than quantitative inputs.

3.2 Data Sources

Multiple heterogeneous datasets were integrated.A multi-source geospatial dataset was constructed to enable grid-based parking intensity modeling. The integrated datasets include:

- Roads and Transport Authority (RTA) Parking Capacity Dataset
- OpenStreetMap (OSM) Road Network (Geofabrik extract)
- OpenStreetMap Points of Interest (POI)
- Dynamic World Land-Cover Dataset (Sentinel-2 derived)

All vector layers were reprojected to EPSG:32640 for Dubai prior to spatial aggregation. Projection consistency ensures that Euclidean distance calculations, overlay operations, and density normalization are expressed in metric units.

3.2.1 Administrative Boundaries

Official Dubai Municipality administrative sector boundaries were used for macro-level reporting and validation. Community-level subdivisions derived from OpenStreetMap enabled secondary aggregation for exploratory spatial interpretation.

3.2.2 Parking Infrastructure Data

Parking supply data were obtained in vector format (KML), containing:

- PARKING ZONE ID
- CAPACITY
- AREA NAME

Capacity values stored as string attributes were converted to numeric format prior to spatial aggregation.

3.2.3 Road Network and Points of Interest

Road network geometries were extracted from OpenStreetMap and classified using the highway attribute. POI datasets were filtered to include activity-relevant urban features (commercial, institutional, and religious categories).

3.2.4 Remote Sensing and Land-Use Data

Land-use composition was derived from the Dynamic World probabilistic land-cover dataset, generated from Sentinel-2 imagery at 10m resolution. Raster-to-vector aggregation was performed via zonal statistics within QGIS, producing land-use proportion variables at grid-cell level. The remote sensing dataset was used for contextual explanatory modeling rather than direct vehicle detection. Police station proximity was incorporated as a structural urban activity proxy. Areas with higher policing infrastructure often correspond to high call-density zones and concentrated activity, which can indirectly influence parking demand intensity.

3.3 Spatial Unit Definition

The inclusion of policy-relevant spatial predictors is informed by official parking management practices in Dubai, where zone-specific tariff regimes and variable pricing influence parking behaviour. Zone designs and designated parking categories (e.g., premium versus standard) directly correlate with observed land-use intensity and accessibility, necessitating features such as commercial density and proximity to major roads in predictive modeling (Roads and Transport Authority (RTA), 2021, 2023d). Furthermore, permit and seasonal parking subscription mechanisms reflect institutional regulation of residential and special-use parking, aligning with the observed spatial heterogeneity in capacity distribution (Roads and Transport Authority (RTA), 2023d).

To mitigate biases associated with irregular administrative geometries, a uniform spatial discretization framework was adopted. Let:

$$\Omega \subset R^2$$

represent the projected Dubai study area in WGS 84 / UTM Zone 40N (EPSG:32640).

A regular tessellation of square grid cells was defined such that:

$$G = \{g_i \mid i = 1, 2, \ldots, n\}$$

where each grid cell g_i has dimensions:

$$250\,\mathrm{m} \times 250\,\mathrm{m}$$

and area:

$$A = 62{,}500\,\mathrm{m}^2.$$

The grid was generated using QGIS grid-creation tools and subsequently clipped to the official Dubai boundary polygon to retain only cells intersecting the study region. Formally,

$$G_{\text{clipped}} = G \cap \Omega.$$

Each grid cell was assigned a unique identifier id_i, forming the primary spatial unit for all feature engineering and modeling operations. The 250 m spatial resolution was selected based on three criteria:

1. Alignment with walkability thresholds (200–400 m pedestrian radius in transport literature),
2. Preservation of intra-district spatial heterogeneity,
3. Avoidance of excessive statistical sparsity.

This discretization reduces the Modifiable Areal Unit Problem (MAUP)(Openshaw, 1984) by enforcing consistent spatial aggregation across heterogeneous land-use zones.

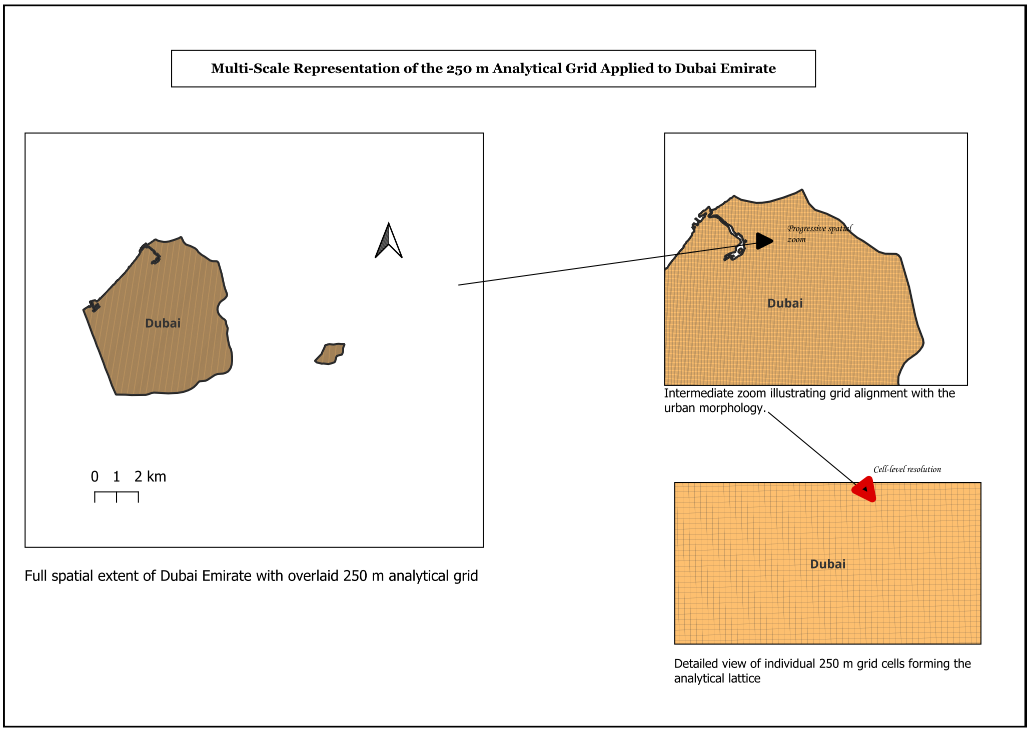


Figure 3.2: 250m analysis grid clipped to Dubai administrative boundary.

3.4 Feature Engineering

All explanatory variables were computed at the grid-cell level. Let

$$\mathbf{X}_i = (x_{i1}, x_{i2}, \ldots, x_{ik})$$

representing the feature vector associated with grid cell g_i. The engineered features include:

- Road density
- Distance to major roads
- Metro accessibility (distance to nearest station)
- Commercial POI density
- Institutional POI density
- Religious POI density
- Residential land-use ratio

- Parking capacity density

All variables were expressed as intensive measures to ensure spatial comparability across grid cells. In general form:

$$\text{Density}_i = \frac{\text{Attribute}_i}{A},$$

where $A = 62{,}500\,\mathrm{m}^2$ represents the area of each grid cell.

3.4.1 Road Infrastructure Intensity

Road geometries were spatially intersected with the grid:

$$R_i = R \cap g_i,$$

where R denotes the full road network dataset. The total road length within grid cell g_i is given by:

$$L_i = \sum_{j=1}^{m_i} \ell_{ij},$$

where ℓ_{ij} represents the length of road segment j intersecting cell i, and m_i is the number of intersecting segments. Road density was therefore computed as:

$$\text{Road Density}_i = \frac{L_i}{A}.$$

Spatial intersection and geometric length calculations were performed using:

- `Geo Pandas` for overlay operations,
- `Shapely` for geometric computations,
- QGIS geometry tools for Preprocessing validation.

3.4.2 Distance to Major Roads

Let c_i denote the centroid of grid cell g_i. The distance to the nearest major road segment is defined as:

$$d_i = \min_{r \in R_{\text{major}}} \|c_i - r\|,$$

where R_{major} represents the subset of major road classes (motorway, trunk, and primary), and $\|\cdot\|$ denotes the Euclidean norm. Distances were computed in projected coordinates (EPSG:32640) to preserve metric accuracy.

3.4.3 Metro Accessibility (Rail Proximity)

Metro accessibility was operationalized as the Euclidean distance from each grid-cell centroid to the nearest active metro station.

Metro station geometries were obtained from OpenStreetMap and filtered to include relevant passenger transport classes (subway, tram, monorail), excluding freight and non-passenger rail infrastructure.

Let $S = \{s_j\}$ denote the set of metro station point geometries and c_i the centroid of grid cell g_i. Metro accessibility was defined as:

$$d_i^{\text{metro}} = \min_{s \in S} \|c_i - s\|,$$

where $\|\cdot\|$ denotes the Euclidean norm in projected coordinate space (EPSG:32640).

Distances were computed in meters after projection to WGS 84 / UTM Zone 40N to ensure metric consistency. Lower values of d_i^{metro} indicate higher accessibility.

This feature captures structural transit accessibility and centrality effects, reflecting the role of metro corridors in concentrating commercial density, pedestrian flows, and parking turnover pressure.

This operationalization aligns with accessibility-based urban modeling frameworks in which transit proximity functions as a structural determinant of activity concentration and land-value gradients (Hansen, 1959).

3.4.4 Points-of-Interest (POI) Density

Points of interest were categorized into demand-relevant classes:

- Commercial
- Institutional

- Religious

For category k, POI density is defined as:

$$\text{POIDensity}_{ik} = \frac{N_{ik}}{A},$$

where N_{ik} is the number of POIs of type k intersecting grid cell g_i. Spatial aggregation was performed using overlay-based counting operations.

3.4.5 Residential Land-Use Ratio

Residential polygons were spatially intersected with grid cells:

$$A_{\text{res},i} = \text{Area}(\text{Residential} \cap g_i).$$

The residential land-use ratio was computed as:

$$\text{ResRatio}_i = \frac{A_{\text{res},i}}{A}.$$

Raster-based zonal statistics and vector overlay operations were conducted in QGIS and verified using Geo Pandas spatial intersections to ensure consistency across processing environments.

3.5 Target Variable Construction

3.5.1 Parking Capacity Aggregation

Parking supply data were obtained from the Roads and Transport Authority (RTA) Open Data Portal in vector (KML) format. The dataset contains spatially referenced parking zones with associated capacity values.

Let P denote the set of parking polygons and g_i the i-th grid cell. Parking capacity was spatially aggregated to the grid level using overlay operations:

$$P_i = P \cap g_i.$$

Total parking capacity per grid cell is defined as:

$$C_i = \sum_{j=1}^{n_i} c_{ij},$$

where c_{ij} represents the capacity of parking zone j intersecting grid cell i, and n_i is the number of intersecting parking zones.

Spatial aggregation was implemented using `Geo Pandas` spatial joins and group-by summation operations, with Preprocessing performed in QGIS to ensure geometric validity.

3.5.2 Capacity Density

To normalize parking supply across spatial units, capacity density was defined as:

$$\text{Capacity Density}_i = \frac{C_i}{A},$$

where $A = 62{,}500\,\text{m}^2$ denotes grid-cell area. This continuous variable represents parking supply intensity per unit area.

3.5.3 Imbalance Index Construction

To capture spatial mismatch between parking supply and urban activity intensity, a normalized imbalance index was constructed.

Let D_i denote a composite demand proxy defined as the aggregated standardized sum of commercial, institutional, and religious POI densities within grid cell g_i.Standardized values were computed using Z-score normalization:

$$Z(X_i) = \frac{X_i - \mu_X}{\sigma_X}.$$

The imbalance index is defined as:

$$\Delta_i = Z(D_i) - Z(C_i).$$

To prevent extreme leverage effects caused by heavy-tailed distributions:

$$\Delta_i^{\text{clipped}} = \text{clip}(\Delta_i, -5, 5).$$

The ± 5 bound corresponds approximately to the extreme tails of the standardized imbalance distribution and prevents undue influence of outliers on spatial autocorrelation statistics. The imbalance index serves as the basis for subsequent spatial autocorrelation testing.

3.6 Machine Learning Modeling Framework

3.6.1 Spatial Autocorrelation Analysis

To evaluate whether parking imbalance exhibits structured spatial clustering, global and local spatial statistics were computed.

3.6.2 Theoretical Framing of the Parking Imbalance Index

The parking imbalance index developed in this study can be interpreted as a reduced-form spatial interaction construct that captures the mismatch between attraction (urban activity intensity) and capacity (parking supply) within standardized neighborhood units.

By aggregating supply to a uniform 250 m grid and approximating activity intensity through POI-derived densities, the index operationalizes a potential-accessibility surface in which opportunity concentration induces localized parking search pressure. This interpretation aligns with classical accessibility formulations in which destinations function as attractors weighted by opportunity magnitude (Hansen, 1959; Wilson, 1967).

The observed spatial clustering of imbalance values—quantified through Global Moran's I and localized Getis–Ord G_i^* statistics—indicates that imbalance is not randomly distributed but exhibits statistically significant spatial dependence (Getis and Ord, 1992; Anselin, 1995). Such dependence is consistent with Tobler's First Law of Geography (Tobler, 1970) and with parking search dynamics in which scarcity or oversupply in one location produces spillover effects in adjacent cells (Tang et al., 2014; Shoup, 2006).

From a spatial econometric perspective, these clusters suggest endogenous feedback mechanisms, where localized parking pressure propagates through near-neighbor substitution and search behavior (Anselin, 1988). The imbalance surface therefore reflects not only descriptive concentration but emergent interaction effects consistent with urban search and competition models.

While the grid framework mitigates distortions associated with irregular administrative boundaries, it introduces sensitivity to spatial scale and weight specification, reflecting the Modifiable Areal Unit Problem (MAUP) (Openshaw, 1984). Accordingly, hotspot classification must be interpreted as inferentially defined spatial regimes contingent on scale and weight assumptions rather than arbitrary zoning categories.

This two-stage framework consists of:

- Inferential hotspot detection via spatial statistics (Moran's I and Getis–Ord G_i^*), followed by
- Supervised regime classification using machine learning.

This structure allows spatial dependence to inform predictive modeling while maintaining transparency regarding statistical assumptions and sensitivity to scale and spatial weights.

Global Moran's I

Global spatial autocorrelation was assessed using Moran's I:

$$I = \frac{n}{W} \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2},$$

where:

- x_i denotes imbalance value at cell i,
- w_{ij} represents elements of the spatial weight matrix,
- $W = \sum_i \sum_j w_{ij}$,
- n is the total number of grid cells.

Spatial weights were constructed using Queen contiguity via the `libpysal` library.

Getis–Ord G_i^* Statistic

Local hotspot detection was performed using the Getis–Ord G_i^* statistic:

$$G_i^* = \frac{\sum_{j=1}^{n} w_{ij} x_j - \bar{X} \sum_{j=1}^{n} w_{ij}}{S \sqrt{\frac{n \sum_{j=1}^{n} w_{ij}^2 - \left(\sum_{j=1}^{n} w_{ij}\right)^2}{n-1}}}.$$

Cells with $G_i^* > 1.96$ were classified as statistically significant hotspots at the 95% confidence level.

The statistically detected hotspot regions were subsequently validated through visual inspection of high-resolution Sentinel-derived imagery retrieved via Google Earth to confirm morphological correspondence between inferred clusters and observable urban density patterns.

3.6.3 Supervised Classification Framework

A supervised machine learning framework was developed to predict the probability of a grid cell being classified as a hotspot. Let:

$$y_i = \begin{cases} 1 & \text{if } G_i^* > 1.96 \\ 0 & \text{otherwise} \end{cases}$$

denote the binary target variable. The feature matrix is defined as:

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \vdots \\ \mathbf{X}_n \end{bmatrix}.$$

Random Forest Model

A Random Forest classifier (Breiman, 2001) was selected for its ensemble-based variance reduction and ability to model nonlinear decision boundaries without strong parametric assumptions.

- Robustness to nonlinear relationships,
- Insensitivity to multicollinearity,

- Stability under class imbalance,
- Built-in feature importance estimation.

The model was implemented using `scikit-learn` with class weighting enabled to mitigate imbalance effects.

Probability threshold calibration was applied:

$$\hat{y}_i = \begin{cases} 1 & \text{if } P(y_i = 1 \mid \mathbf{X}_i) > \tau \\ 0 & \text{otherwise} \end{cases}$$

where τ was later tuned between 0.7 and 0.8 to balance precision and recall.

3.7 Cross-City Generalization Assessment

To evaluate spatial transferability, the Dubai-trained Random Forest model was applied to an out-of-sample urban domain in Seattle, United States.

All Seattle datasets were reprojected into WGS 84 / UTM Zone 10N (EPSG:32610), the appropriate projected coordinate reference system for the Pacific Northwest longitudinal zone. This ensured metric consistency in distance-based computations and area normalization.

An identical $250\,\text{m} \times 250\,\text{m}$ grid was generated over the Seattle study region. Feature engineering procedures were replicated without modification, including:

- Road density computation,
- Euclidean distance to major roads,
- POI density aggregation,
- Residential land-use ratio calculation,
- Parking capacity density normalization.
- Metro accessibility (distance to nearest station),

No retraining or recalibration was performed using Seattle data.Feature scaling parameters derived from the Dubai training dataset were preserved during transfer to avoid information leakage. The trained Dubai model was directly applied to the Seattle feature matrix:

$$\hat{y}_i^{(SEA)} = f_{\text{Dubai}}(\mathbf{X}_i^{(SEA)}).$$

Model predictions were compared against locally computed Gi* hotspot classifications in Seattle to evaluate predictive robustness under spatial domain shift. This cross-city experiment assesses structural transferability of spatial determinants rather than universal predictive validity.

3.8 Limitations

Despite the structured spatial-statistical modeling framework, several limitations must be acknowledged.

First, the study relies on infrastructure-based parking capacity data rather than real-time occupancy or transaction-level usage measurements. Consequently, the model captures structural parking intensity rather than dynamic behavioral demand patterns.

Second, land-use proportions derived from the Dynamic World dataset represent probabilistic classifications from Sentinel-2 imagery at 10 m resolution. While suitable for urban-scale modeling, this resolution does not allow detection of individual vehicles or precise occupancy estimation.

Third, the 250 m grid resolution introduces spatial aggregation effects. Although selected to balance granularity and statistical stability, alternative discretizations may yield different clustering sensitivities.

Fourth, spatial statistical results depend on the specification of spatial weights. Queen contiguity was adopted for robustness; however, distance-based kernels or k-nearest-neighbor structures may produce variation in hotspot delineation.

Fifth, cross-city generalization assumes partial structural similarity between Dubai and Seattle. Differences in zoning policy, socio-economic structure, and parking regulation may influence predictive stability.

Finally, although spatial features were incorporated, the Random Forest model does not explicitly model spatial dependence within its algorithmic structure. Residual spatial autocorrelation may therefore remain.

3.9 Implementation Framework: Integrated GIS–Python Workflow

The analytical pipeline was implemented through an integrated GIS–Python framework combining QGIS Preprocessing and Python-based statistical modeling.

3.9.1 Software Environment

The computational environment consisted of:

- QGIS (Long-Term Release) for spatial Preprocessing and grid generation,
- PyQGIS for automation of spatial operations,
- Python 3.x scientific stack for statistical modeling.

Primary Python libraries included:

- `Geo Pandas` for vector spatial operations,
- `Shapely` for geometric computations,
- `NumPy` for numerical operations,
- `Pandas` for tabular manipulation,
- `libpysal` for spatial weight matrix construction,
- `esda` for Moran's I and Getis–Ord G_i^* statistics,
- `scikit-learn` for supervised machine learning,
- `Matplotlib` for visualization.

3.9.2 Spatial Processing Workflow

Spatial Preprocessing was conducted in QGIS and partially automated using PyQGIS scripts.

Core steps included:

1. Import administrative boundaries and reproject to appropriate UTM zone.
2. Generate 250 m grid tessellation.
3. Clip grid to study boundary.
4. Perform overlay operations for roads, POIs, and parking polygons.
5. Compute zonal statistics for land-use rasters.

Vector aggregation and feature normalization were subsequently verified in Python using Geo Pandas.

3.9.3 Spatial Statistics Implementation

Spatial weights were constructed using Queen contiguity:

$$w_{ij} = \begin{cases} 1 & \text{if } g_i \text{ shares a boundary with } g_j \\ 0 & \text{otherwise} \end{cases}$$

Weights were row-standardized prior to Moran's I and Gi* computation using `libpysal`.

3.9.4 Machine Learning Implementation

The Random Forest classifier was implemented using `scikit-learn`. Hyper parameters included:

- Number of trees: 300–600,
- Maximum depth: 10–20,
- Class weighting: balanced,
- Probability threshold tuning: $\tau \in [0.7, 0.8]$.

Model training and validation were performed using stratified sampling to preserve hotspot class proportions.

3.9.5 Reproducibility and Export

All intermediate outputs were stored in Geo Package format to preserve spatial integrity. Final outputs included:

- Geo Package spatial layers,
- CSV feature matrices,
- Serialized Random Forest model (pickle format).

The integrated workflow ensures methodological transparency, reproducibility, and adaptability to additional urban contexts.

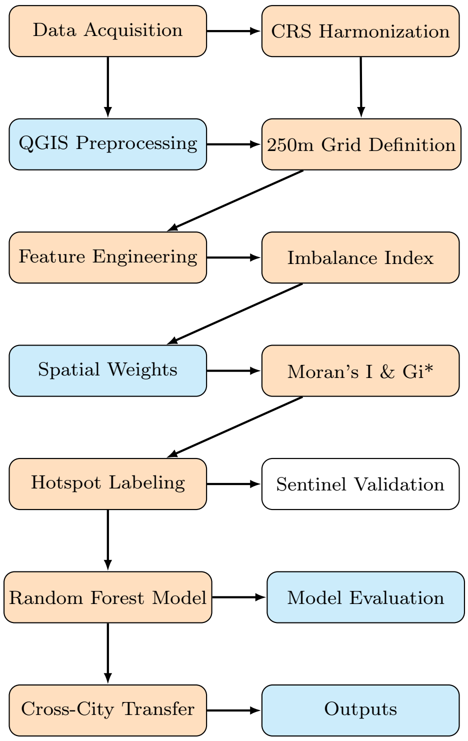


Figure 3.3: Compact representation of the integrated GIS–AI analytical workflow.

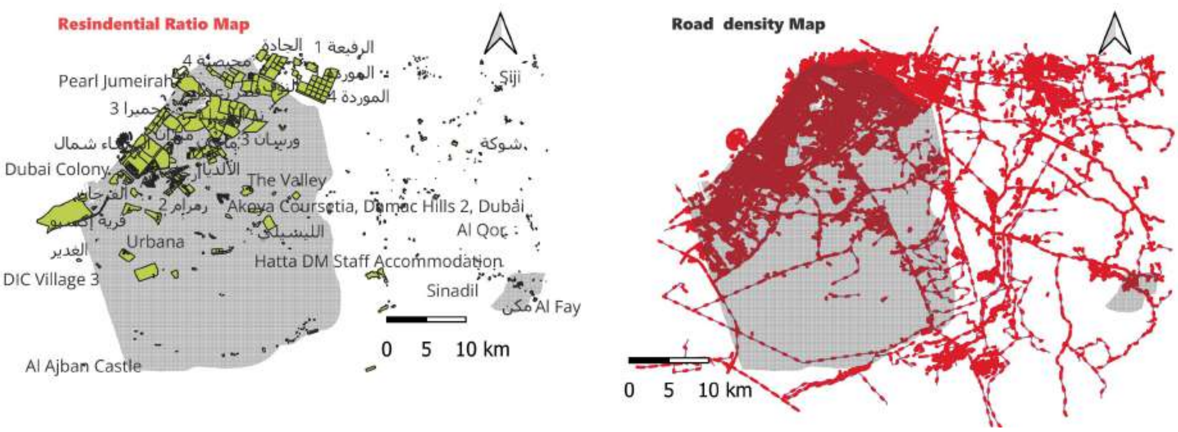
Chapter 4

Results and Analysis

4.1 Descriptive Spatial Structure

The 250 m grid framework produced 152,322 spatial cells across the Dubai study area. Of these, 1,148 cells contained non-zero parking capacity values, indicating strong spatial concentration of formal parking supply infrastructure.

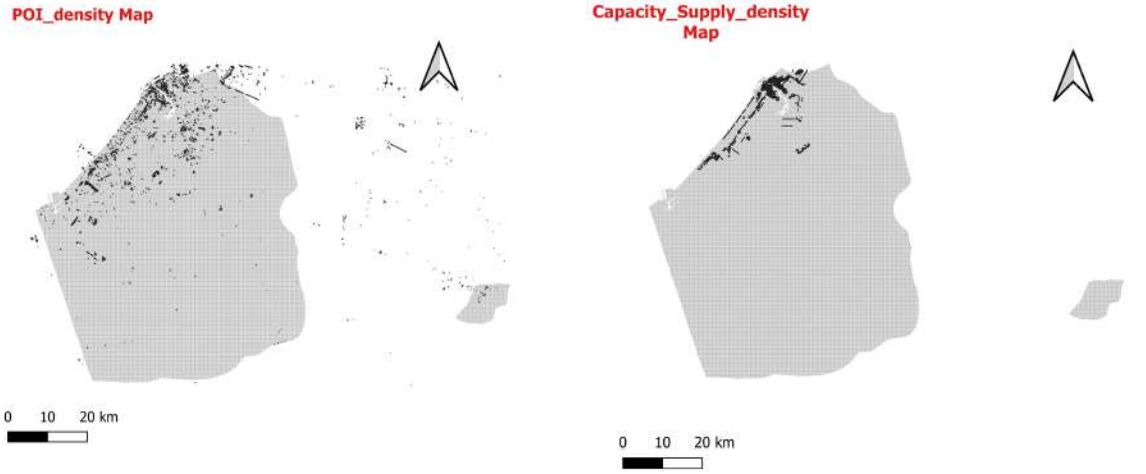
The mean POI count per grid cell was 0.09, reflecting spatial sparsity outside major commercial corridors. The mean parking capacity among active cells was 356 spaces, confirming the presence of large centralized supply clusters rather than uniformly distributed micro-parking infrastructure.



(a) Residential land-use ratio per grid cell.

(b) Road density per 250 m grid cell.

Figure 4.1: Spatial distribution of residential intensity and road network density across the 250 m grid framework.



(a) POI density per grid cell.

(b) Parking capacity density per 250 m grid cell.

Figure 4.2: Spatial comparison of activity intensity and parking supply distribution across the 250 m grid framework.

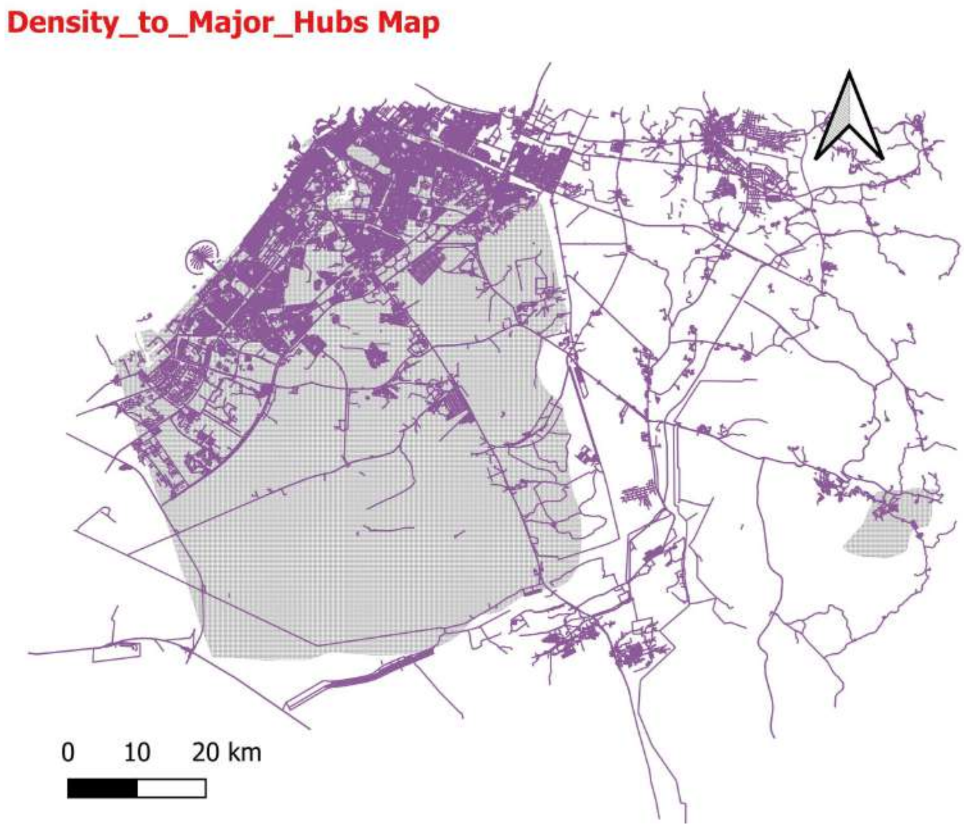


Figure 4.3: Distance from grid centroids to nearest major road.

4.2 Feature Diagnostics and Multicollinearity Assessment

Before supervised modeling, feature interdependence was evaluated to prevent redundancy bias and ensure model stability.

Pearson correlation coefficients were computed across engineered spatial predictors using the following procedure:

```python
import seaborn as sns
import matplotlib.pyplot as plt
# Compute correlation matrix
corr_matrix = df[feature_columns].corr(method="pearson")
# Plot heatmap
plt.figure(figsize=(10, 8))
sns.heatmap(
    corr_matrix,
    annot=True,
    cmap="coolwarm",
    fmt=".2f",
    linewidths=0.5
)
plt.title("Feature Correlation Matrix")
plt.tight_layout()
plt.show()
```

Listing 4.1: Computation of Pearson correlation matrix for engineered spatial predictors

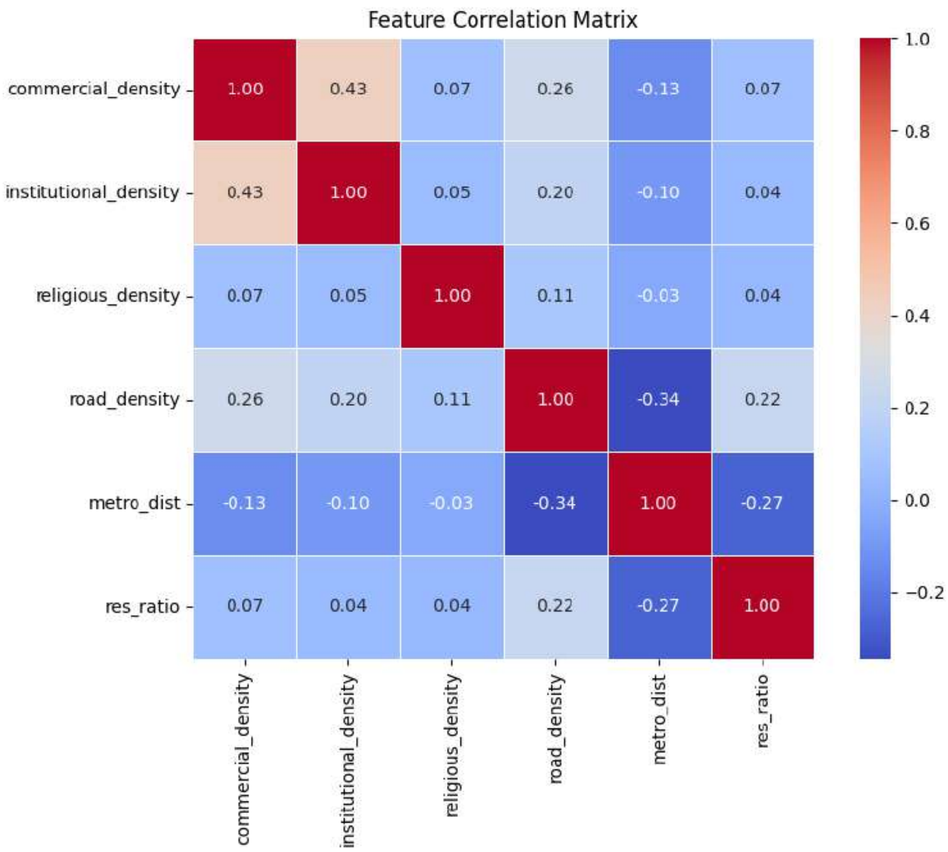


Figure 4.4: Feature correlation matrix for engineered spatial predictors.

The correlation matrix indicates no severe multicollinearity among predictors, as all pairwise Pearson coefficients remain well below the threshold of $|r| > 0.85$. The strongest observed association ($r = 0.43$) occurs between commercial and institutional density, reflecting expected co-location patterns within mixed-use urban cores. A moderate negative correlation between road density and metro distance ($r = -0.34$) confirms the accessibility gradient characteristic of Dubai's spatial structure. Overall, the predictor set demonstrates acceptable independence for ensemble modeling.

Feature Engineering Justification: POI as Demand Proxy

Points-of-interest (POI) density was incorporated as a structural demand proxy for the following reasons:

- POIs spatially approximate concentrations of economic and social activity,
- Commercial and institutional density correlates with parking turnover intensity,
- Religious and civic nodes generate periodic parking surges.

This feature design directly addresses **Research Question 1**:

> *What spatial determinants explain structural parking imbalance in Dubai?*

The strong clustering observed in POI-dense northern coastal zones confirms their explanatory relevance within the broader parking imbalance framework.

4.3 Parking Imbalance Index

To quantify spatial mismatch between parking supply and demand intensity, a normalized imbalance index was constructed as:

$$\Delta_i = Z(D_i) - Z(C_i),$$

where D_i represents the demand proxy (POI density) and C_i denotes parking capacity density within grid cell i. The operator $Z(\cdot)$ denotes standard score normalization.

Standardization ensures that both variables are placed on a comparable scale prior to differencing. Positive values of Δ_i indicate relative under-supply of parking, while negative values indicate relative oversupply.

Preliminary inspection of the spatial distribution of Δ_i suggests visible concentration of higher imbalance values along the northern coastal corridor. However, visual patterns alone do not provide statistical confirmation of clustering. Formal spatial autocorrelation testing is therefore conducted in the following section

4.4 Spatial Autocorrelation Analysis

4.4.1 Global Moran's I

Global spatial autocorrelation of the imbalance index was evaluated using Moran's I statistic.

```python
from esda.moran import Moran
from libpysal.weights import Queen

# Construct Queen contiguity spatial weights
w = Queen.from_dataframe(gdf)
w.transform = "r"  # row-standardized weights

# Compute Global Moran's I
moran = Moran(gdf["imbalance"], w)

# Report statistic and p-values
print("Moran's I:", moran.I)
print("p-value (normal approximation):", moran.p_norm)
print("p-value (permutation):", moran.p_sim)
```

Listing 4.2: Computation of Global Moran's I for the parking imbalance index

The computed Moran's I value was:

$$I = 0.7037, \quad p < 0.001.$$

The high positive Moran's I value indicates strong positive spatial autocorrelation. Grid cells exhibiting high imbalance values are surrounded by similarly high values, while low values cluster with low values. The statistically significant p-value rejects the null hypothesis of spatial randomness.

This result confirms that parking imbalance is spatially structured rather than randomly distributed across the study area, directly addressing Research Question 2

4.4.2 Local Hotspot Detection (Getis–Ord G_i^*)

To identify the exact spatial locations of statistically significant clusters, the Getis–Ord G_i^* statistic was computed.

```python
from esda.getisord import G_Local
from libpysal.weights import Queen

# Construct Queen contiguity spatial weights
w = Queen.from_dataframe(gdf)
w.transform = "r"

# Compute Gi* z-scores
gi = G_Local(gdf["imbalance"], w)
gdf["GiZ"] = gi.Zs
```

Listing 4.3: Computation of Getis–Ord G_i^* statistic

Hotspots were defined using the threshold:

$$G_i^* > 1.96,$$

corresponding to a 95% confidence level.

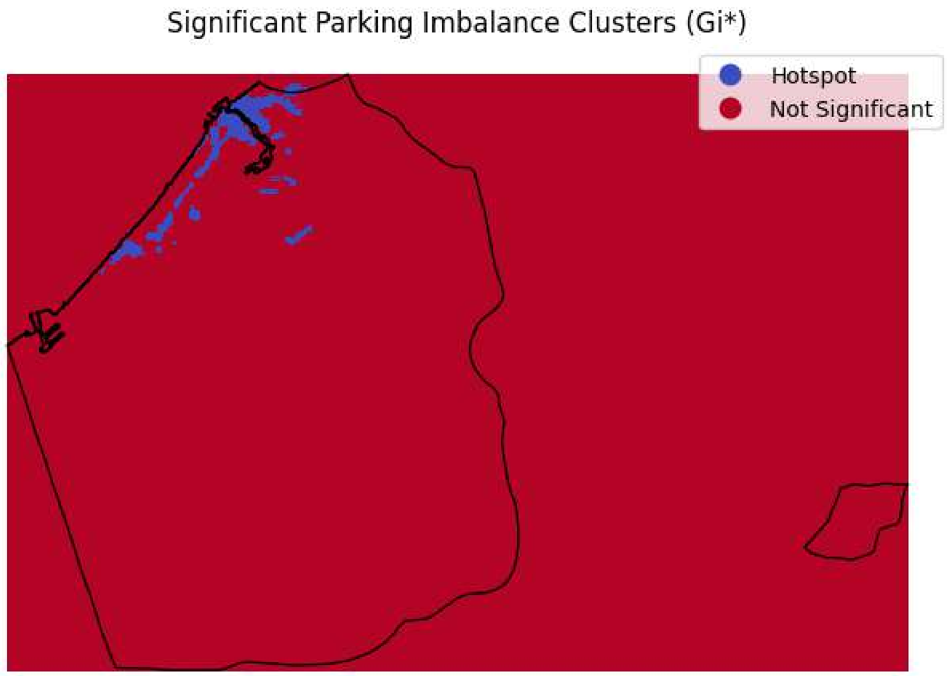


Figure 4.5: Statistically significant parking imbalance hotspots identified using Getis–Ord G_i^*.

The local hotspot analysis reveals statistically significant clustering concentrated along the northern coastal corridor. These hotspot regions align with high-density commercial and mixed-use districts characterized by elevated accessibility and activity intensity.

The remainder of the study area is largely classified as non-significant, indicating that extreme imbalance conditions are spatially localized rather than uniformly distributed.

Together, the global and local spatial statistics demonstrate that parking imbalance follows a clear urban core–periphery structure, with deficit pressure concentrated in the economic core. This provides strong inferential evidence that parking imbalance is spatially clustered rather than randomly distributed.

The spatial clustering of parking imbalance aligns with RTA pricing and zoning practices, where premium commercial districts (e.g., CBD, Marina, Business Bay) impose higher parking tariffs and greater demand turnover (Roads and Transport Authority (RTA), 2023d; Shoup, 2006).

Conversely, peripheral residential zones often exhibit lower demand intensity and oversupply conditions, consistent with relatively lower tariff structures and regulatory emphasis on residential permits (Roads and Transport Authority (RTA), 2023d).

These findings suggest that observed structural imbalance is not merely a statistical artifact but reflects policy-driven economic signals embedded within Dubai's parking system.

4.5 Supervised Modeling Results (Dubai)

Following feature engineering and hotspot label construction via Getis–Ord G_i^*, a Random Forest classifier was implemented to predict statistically significant imbalance zones. Three configurations were evaluated:

- Baseline Random Forest
- Threshold-adjusted classifier ($\tau = 0.7$)
- Strengthened model (600 trees, depth = 20)

The strengthened configuration achieved the most stable performance.

Table 4.1: Comparison of Random Forest Model Configurations

Model	Accuracy	Hotspot Precision	Hotspot Recall
Baseline RF	0.965	0.16	0.90
Threshold Adjusted ($\tau = 0.7$)	0.948	0.38	0.68
Strengthened RF (Final)	0.944	0.67	0.48

While overall accuracy slightly decreased in the strengthened model, hotspot precision improved substantially, reducing false positives in high-demand planning zones. This configuration was selected as the operational Dubai model.

4.5.1 Feature Importance

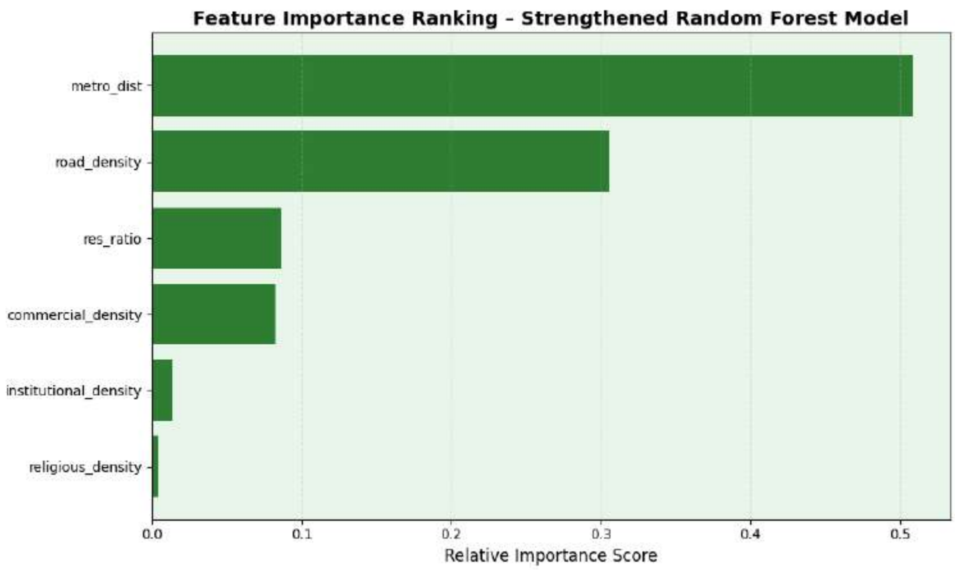


Figure 4.6: Feature importance ranking from strengthened Random Forest model.

Commercial density and metro accessibility emerged as the dominant explanatory variables, followed by road density and institutional density. Religious density and residential ratio demonstrated secondary but spatially coherent influence.

This confirms that structural parking imbalance is strongly associated with economic intensity and mobility centrality.

4.6 Morphological Validation Using Sentinel Imagery

To validate statistically identified hotspot clusters, high-resolution Sentinel imagery (retrieved via Google Earth interface) was visually inspected for selected high-confidence hotspot cells.

(a) Deira hotspot cluster (CBD coastal corridor).

(b) Dubai Marina hotspot cluster.

Figure 4.7: Sentinel-based morphological validation of statistically detected hotspot zones. Both locations exhibit dense commercial structures and concentrated parking infrastructure consistent with high G_i^* clustering.

Hotspot Regions (High Imbalance)	**Coldspot Regions (Oversupply / Low Demand)**
Dense commercial blocks	Low-density peripheral housing
Large parking structures	Industrial buffer zones
Transit-adjacent activity nodes	Limited economic and social clustering
High accessibility corridors	Reduced network centrality
Mixed-use vertical development	Dispersed land-use patterns

Table 4.2: Morphological comparison between statistically significant hotspot and coldspot regions derived from Getis–Ord G_i^* analysis and Sentinel imagery validation.

This qualitative validation confirms morphological consistency between statistical clustering and observable urban structure.

4.6.1 Confusion Matrix Analysis

Table 4.3: Confusion Matrix for Strengthened Random Forest Model (Dubai)

	Predicted Non-Hotspot	Predicted Hotspot
Actual Non-Hotspot	141,882	315
Actual Hotspot	602	523

The model achieved an Area Under the Curve (AUC) of 0.94, indicating strong separability between hotspot and non-hotspot regimes.

Integration Point: From Prediction to Policy Interpretation After model training and calibration, the output of the supervised model is a grid-level probability surface $\hat{p}_i$ indicating the likelihood that each cell belongs to a statistically significant hotspot regime. These probabilities and their associated explanatory signals (e.g., top contributing features for each cell) are then packaged into a structured "decision payload" that becomes the input to the RAG layer. In this design, predictive modeling remains strictly separated from textual reasoning: the Random Forest produces the classification outputs, while RAG retrieves relevant planning and regulatory context (e.g., RTA guidance, zoning codes, TOD strategies) and synthesizes policy-aligned explanations grounded in retrieved sources. This ensures the interpretability layer augments decision support without modifying or re-optimizing the underlying classifier.

4.7 Cross-City Transferability Assessment

The strengthened Dubai model was applied to a Seattle test region without retraining.

Prediction probabilities exhibited near-uniform low values across the Seattle grid, indicating limited structural generalization.

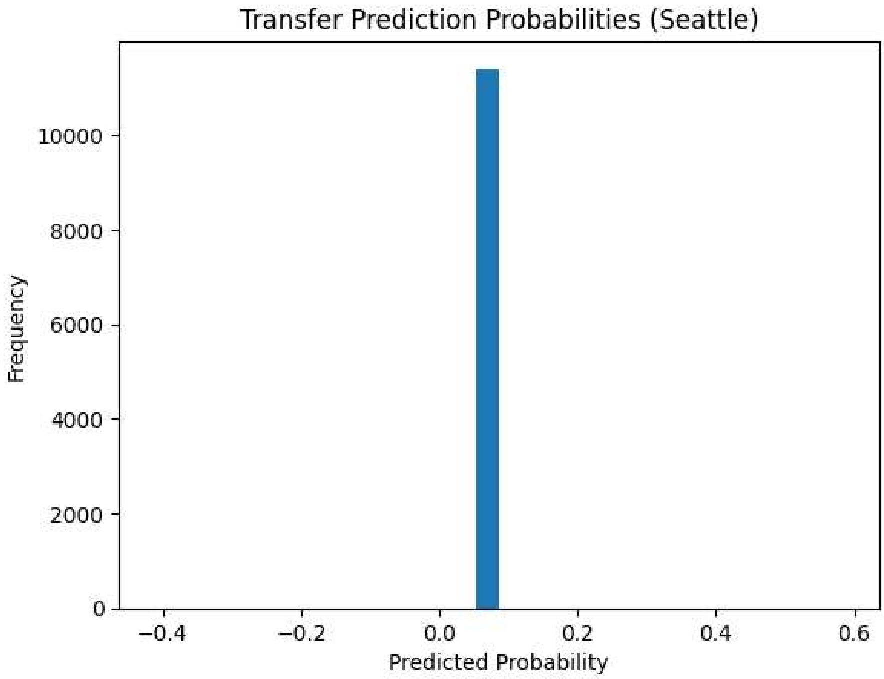


Figure 4.8: Probability distribution of Dubai-trained model applied to Seattle.

The compressed probability distribution suggests contextual sensitivity of learned relationships.

This result demonstrates spatial non-stationarity: parking imbalance determinants are not universally transferable across distinct urban morphologies without recalibration.

4.7.1 Spatial Non-Stationarity and Domain Sensitivity

The limited generalization performance observed when transferring the Dubai-trained model to Seattle reflects the principle of spatial non-stationarity. Spatial non-stationarity implies that relationships between predictors and outcomes vary across geographic contexts, rather than remaining globally constant. In the context of parking systems, determinants of structural imbalance are shaped by urban morphology, regulatory regimes, accessibility hierarchies, and land-use organization.

Dubai exhibits a highly centralized coastal development corridor characterized by vertical mixed-use clusters, large structured parking facilities, and strong metro-linked accessibility gradients. Commercial intensity is spatially concentrated along the northern coastal axis, generating pronounced imbalance clusters detectable through spatial autocorrelation statistics. In contrast, Seattle's urban structure reflects a more polycentric morphology, lower average density gradients, and a historically incremental zoning evolution. Such structural differences reduce the direct transferability of learned decision boundaries derived from Dubai's spatial feature distributions.

Zoning and land-use regulation further differentiate the two contexts. Dubai's development model emphasizes centralized commercial agglomeration and high-capacity parking provision within designated activity corridors. Seattle, by comparison, operates under mature zoning ordinances that incorporate mixed-use neighborhood planning, parking minimum reforms, and localized land-use controls. These institutional differences alter the spatial coupling between activity intensity and parking supply, thereby modifying the statistical relationships captured during model training.

Parking regulation regimes also diverge significantly. Theoretical work on cruising behavior demonstrates that pricing structures and curb management policies directly influence parking search dynamics and spatial spillovers (Shoup, 2006). Where curb pricing approaches market equilibrium, cruising externalities decline, reducing concentrated imbalance effects. Differences in enforcement intensity, permit systems, and off-street pricing policies therefore reshape the spatial manifestation of parking pressure across cities.

Moreover, parking search processes are inherently network-dependent and influenced by localized substitution behavior (Tang et al., 2014). When drivers respond differently to scarcity signals due to regulatory, cultural, or network structure differences, the emergent spatial clustering patterns shift accordingly. This reinforces the argument that parking imbalance operates as a context-sensitive spatial interaction process rather than a universally stable structural pattern.

From a broader spatial econometric perspective, these findings align with the notion that spatial processes are scale- and domain-contingent (Anselin, 1988). The Modifiable Areal Unit Problem (MAUP) further implies that aggregation structure and zoning logic influence observed clustering regimes (Openshaw, 1984). Consequently, predictive relationships learned in one metropolitan morphology cannot be assumed invariant under cross-domain transfer.

Importantly, the observed transfer limitation does not indicate model inadequacy. Rather, it empirically confirms that parking imbalance is an emergent phenomenon shaped by localized accessibility gradients, regulatory frameworks, and urban form. The Dubai-trained model successfully captured the structural determinants specific to its training environment. However, the Seattle application demonstrates that parking systems exhibit domain sensitivity, requiring contextual recalibration or domain adaptation techniques for robust cross-city deployment.

4.8 Retrieval-Augmented Generation (RAG) Conceptual Integration

To enhance interpretability and governance applicability, a conceptual Retrieval-Augmented Generation (RAG) layer was designed to sit atop the predictive modeling framework. The RAG layer was conceptually designed and not deployed in a production environment during the present study. RAG systems combine:

- Vector-based document retrieval,
- Large language model synthesis,
- Structured data embeddings.

Within the proposed architecture:

1. The Random Forest model generates hotspot probabilities.
2. Grid-level feature vectors are embedded as structured queries.
3. Relevant planning documents (RTA regulations, zoning codes,mobility strategies) are retrieved from a knowledge base.
4. A language model synthesizes contextual policy explanations.

Formally:

$$\text{payload}_i = \{\text{id}_i, \hat{p}_i, \hat{y}_i, \text{TopFeatures}_i, \text{Location}_i\}, \qquad \mathbf{e}_i = f(\text{payload}_i), \qquad \text{Docs}_i = \text{TopK}(\mathbf{e}_i),$$

$$\text{PlanningInsight}_i = \text{LLM}(\text{Docs}_i, \text{payload}_i).$$

The RAG layer does not alter classification outputs; rather, it provides semantic contextualization and policy-aligned interpretation. This approach ensures transparency by separating:

- Statistical detection,
- Predictive modeling,
- Knowledge retrieval,
- Textual reasoning.

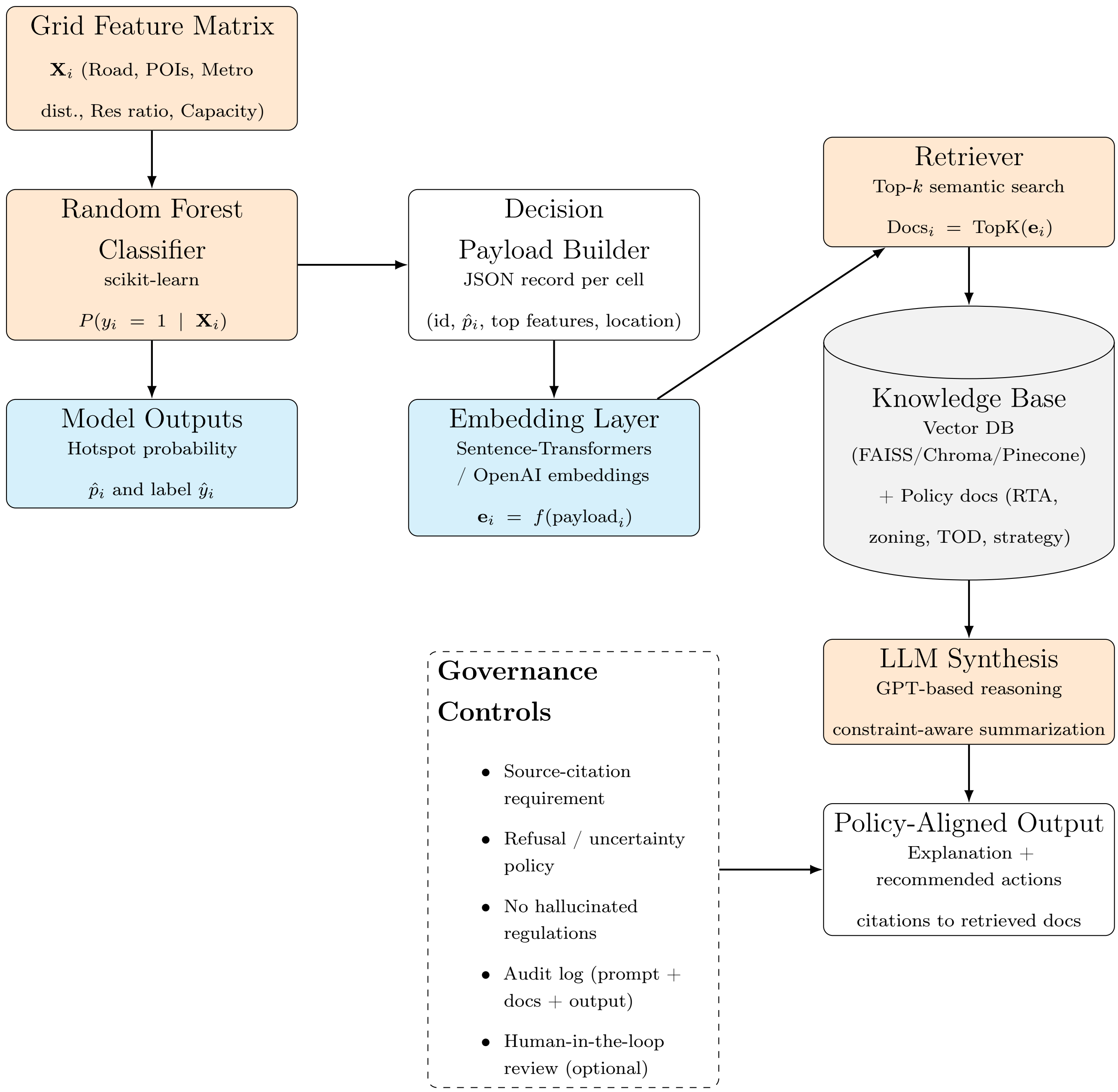


Figure 4.9: Conceptual Retrieval-Augmented Generation (RAG) layer operating as a post-model interpretation and governance component. The predictive model produces hotspot probabilities, which are packaged into a structured query payload, embedded, matched against a policy knowledge base, and synthesized into evidence-linked planning guidance under explicit governance controls.

Chapter 5

Discussion

5.1 Interpretation of Spatial Structure

This chapter interprets the empirical findings in relation to urban parking behavior and decision-making in Dubai. While the analytical implementation of the study is grounded in GIS, spatial statistics, and machine learning, the broader contribution lies in showing how these methods can be integrated within a unified decision-support perspective to interpret parking-related pressures, accessibility constraints, and governance-relevant intervention needs.

Significant Getis–Ord G_i^* hotspot clusters were identified in the northern coastal corridor, corresponding to high-density commercial, institutional, and mixed-use districts. These areas exhibit persistent structural imbalance, characterized by elevated demand proxies relative to available parking capacity.

Conversely, peripheral districts demonstrate statistically significant coldspots, indicating spatial oversupply relative to observed urban activity intensity.

This spatial pattern aligns with classical urban economic theory, where accessibility gradients, commercial clustering, and transport centrality concentrate demand within economically dominant corridors.

Taken together, these patterns indicate that parking imbalance in Dubai should be interpreted not only as a spatial allocation issue, but also as a manifestation of urban parking behavior and parking-related decision-making under uneven accessibility and supply conditions.

5.2 Answering the Research Questions

Research Question 1: What spatial determinants explain structural parking imbalance in Dubai?

Results demonstrate that imbalance is primarily driven by economic intensity and accessibility centrality. Commercial and institutional density strongly correlate with hotspot formation, while metro proximity amplifies clustering effects.In this sense, the identified hotspot structure reflects aggregated decision environments in which users respond to concentrated opportunity fields, constrained supply, and accessibility gradients.

Research Question 2: Can supervised learning accurately predict imbalance zones?

Yes. The strengthened Random Forest classifier achieved high hotspot precision (0.67) with controlled false positive rates, confirming that engineered spatial features contain sufficient structural signal.This predictive capability is important not only for spatial classification, but also for understanding where parking-related decision pressures are most likely to emerge within the urban system.

The statistically significant hotspots located within Dubai's central commercial corridors corroborate the RTA's demand management strategy of zone-based pricing, which seeks to encourage efficient parking turnover and reduce congestion in high-activity zones (Dubai Government, 2022; Roads and Transport Authority (RTA), 2023d).

However, the resultant structural imbalance highlights potential equity considerations, as pricing signals concentrate parking pressure in areas with high economic activity. Policy levers such as dynamic tariffs, targeted permit reforms, and integration with public transit incentives may ameliorate these pressures by redistributing demand across spatially differentiated regulatory conditions (Roads and Transport Authority (RTA), 2023d).

The detected oversupply coldspots further emphasize opportunities for adaptive reuse or policy recalibration to align infrastructure with observed mobility needs.

Research Question 3: Are imbalance determinants transferable across cities?

Transfer testing suggests limited generalization. The Dubai-trained model failed to identify high-confidence hotspots in Seattle, indicating that parking imbalance drivers are context-specific and dependent on localized morphology.This further suggests that parking behavior and decision-making are spatially conditioned by local morphology, regulation, and accessibility structure, rather than being universally transferable across urban contexts.

5.3 Role of Engineered Features

Feature diagnostics and multicollinearity analysis confirmed the statistical stability of the engineered predictors. No severe collinearity ($|r| > 0.85$) was detected, supporting independent explanatory contribution across variables. Beyond statistical robustness, the explanatory variables reflect observable patterns in Dubai's urban life and mobility structure.

5.3.1 POI Density as a Structural Demand Proxy

Points-of-interest density emerged as the most influential predictor of hotspot classification, particularly commercial and institutional categories. This finding aligns with classical accessibility theory, where destinations function as attractors weighted by opportunity magnitude (Hansen, 1959). In practice, Dubai's economic geography is characterized by concentrated activity nodes large shopping malls, business districts, mixed-use towers, religious centers, and institutional campuses where employment and leisure functions co-locate vertically and horizontally.

Residents and commuters in Dubai commonly travel to centralized commercial corridors such as Deira, Business Bay, Dubai Marina, and Downtown. These clusters concentrate employment, retail, and service functions within relatively compact spatial envelopes. Such agglomeration produces predictable parking demand surges, particularly during working hours, weekend retail peaks, and prayer times near mosques. The model's identification of POI density as a dominant predictor therefore reflects not merely statistical correlation but the lived intensity of Dubai's centralized economic structure.

This observation is consistent with spatial spillover theory and search dynamics in parking systems (Tang et al., 2014). As activity density increases, drivers engage in localized search behavior when supply constraints are encountered, producing clustered imbalance effects detectable through Getis–Ord G_i^* statistics.

5.3.2 Road Density and Accessibility Centrality

Road density and proximity to major corridors also ranked highly in feature importance. Dubai's urban fabric is strongly structured around arterial road hierarchies and expressways, shaping accessibility gradients across the metropolitan area. High road density in central districts corresponds with high vehicular throughput and concentrated activity, reinforcing the accessibility–parking coupling identified in the model.

From a spatial econometric perspective, this aligns with the principle that spatial interaction intensity increases with network connectivity (Anselin, 1988). Areas with dense road infrastructure facilitate higher inflows of vehicles, thereby increasing the likelihood of imbalance when parking capacity does not scale proportionally.

Moreover, Dubai's car-oriented development trajectory amplifies this relationship. While the metro system provides major transit corridors, first- and last-mile dependency remains partially vehicle-based in many districts. This hybrid mobility structure strengthens the predictive value of accessibility variables in hotspot detection.

5.3.3 Residential Ratio and Demand Stabilization

The residential land-use ratio exhibited comparatively weaker predictive strength, yet its spatial interpretation remains meaningful. Purely residential neighborhoods in Dubai often contain structured or allocated parking provisions, reducing variability in demand intensity. Unlike commercial districts, residential areas display more temporally stable occupancy patterns, particularly outside peak commuting windows.

This pattern resonates with parking equilibrium theory, where concentrated economic nodes generate higher volatility and search externalities (Shoup, 2006). In contrast, residential clusters experience more predictable demand, reducing the formation of statistically significant imbalance clusters.

5.3.4 Engineered Features as Reflections of Urban Experience

Collectively, the engineered predictors capture fundamental dimensions of Dubai's urban experience:

- Economic concentration (POI density),

- Network accessibility (road density and major road proximity),
- Functional land-use composition (residential ratio),
- Structured supply intensity (capacity density).

The alignment between model-derived feature importance and observable urban behavior reinforces the construct validity of the feature engineering framework. Rather than abstract statistical inputs, the predictors operationalize tangible aspects of daily life in Dubai: commuting to centralized business districts, weekend retail surges in mixed-use developments, and differential parking stability between residential and commercial zones.

This convergence between empirical results, lived urban morphology, and theoretical accessibility formulations strengthens the interpretive credibility of the model. It demonstrates that the supervised classifier is not identifying arbitrary statistical artifacts, but systematically learning the spatial logic embedded within Dubai's mobility and land-use system.

5.3.5 Urban Form, Accessibility and Car Dependency

The dominance of commercial density, institutional activity, and accessibility-related variables within the Random Forest model reflects deeper structural characteristics of Dubai's urban system. Rather than identifying isolated parking shortages, the classifier appears to capture the spatial imprint of opportunity concentration and automobile-oriented mobility patterns.

Accessibility theory posits that mobility flows intensify toward concentrated opportunity fields (Hansen, 1959). In Dubai, retail, employment, and leisure functions are frequently centralized within high-intensity mixed-use clusters and large mall complexes. These developments embed structured parking within vertical podiums and multi-level garages, yet the aggregation of activity within limited spatial envelopes generates concentrated vehicular inflows. The resulting demand peaks are not evenly distributed but spatially clustered along major accessibility corridors.

Road density and proximity to primary arterials therefore emerge as meaningful predictors not merely because of infrastructure presence, but because they proxy the network centrality through which demand is funneled. Spatial econometric theory emphasizes that such concentrated inflows generate endogenous feedback effects, where localized scarcity induces spillovers into adjacent cells (Anselin, 1988). The significant hotspot clusters detected via Getis–Ord G_i^* reflect these self-reinforcing interaction processes.

Dubai's broader car dependency further strengthens this coupling. Despite investment in metro corridors, private vehicle use remains a primary access mode for many residential and commercial trips. In highly motorized environments, limited modal substitution intensifies the translation of accessibility gradients into parking pressure. When alternative transport options are less competitive in first- and last-mile contexts, concentrated opportunity nodes manifest as concentrated parking imbalance surfaces.

This structural dependence helps explain why the supervised model achieved strong separability between hotspot and non-hotspot cells. The engineered predictors encode stable urban morphology signals rather than short-term behavioral noise. In this sense, the model indirectly maps the spatial footprint of car-oriented development and centralized retail patterns, reinforcing theoretical arguments regarding cruising behavior and congestion externalities (Shoup, 2006; Tang et al., 2014).

The convergence between model-derived feature importance, spatial clustering statistics, and observable urban mobility structure strengthens the interpretive validity of the analytical framework. Parking imbalance in Dubai is therefore best understood as an emergent outcome of accessibility concentration, governance regimes, and modal reliance embedded within the metropolitan system.

5.4 Machine Learning Performance Interpretation

The supervised classification framework was designed to predict statistically significant parking imbalance hotspots derived from the Getis–Ord G_i^*statistic.

Three Random Forest configurations were evaluated, reflecting incremental calibration and structural refinement.

5.4.1 Model Accuracy: Mathematical and Practical Interpretation

Accuracy is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where:

- TP = True Positives (correctly predicted hotspots)
- TN = True Negatives (correctly predicted non-hotspots)
- FP = False Positives
- FN = False Negatives

The strengthened Random Forest model achieved an overall accuracy of 0.944. Mathematically, this indicates that approximately 94.4% of all grid cells were correctly classified.

However, in spatial classification problems with imbalanced classes where non-hotspot cells dominate accuracy alone can be misleading. Because hotspot cells constitute a minority class, a naive classifier predicting all cells as non-hotspot would still achieve high apparent accuracy.

Therefore, additional metrics are necessary to evaluate model validity in policy-relevant contexts.

5.4.2 Precision and Recall: Interpretation in Spatial Planning Context

Precision is defined as:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall is defined as:

$$\text{Recall} = \frac{TP}{TP + FN}$$

In hotspot detection:

- Precision reflects how many predicted hotspots are truly significant.
- Recall reflects how many true hotspots were successfully identified.

The strengthened model achieved:

- Precision = 0.67

- Recall = 0.48

This indicates a deliberate trade-off. By adjusting the probability threshold to $\tau = 0.7$, false positives were reduced substantially. In planning terms, this minimizes the risk of overestimating deficit zones and misallocating intervention resources.

Lower recall suggests some true hotspots were not detected; however, from a governance perspective, high precision is often preferable. Overestimating deficit zones may lead to inefficient capital allocation, whereas missing marginal hotspots can be addressed through periodic recalibration.

5.4.3 Model Configuration Comparison

Table 5.1: Comparative Performance of Random Forest Configurations

Model	Accuracy	Precision	Recall
Baseline RF	0.965	0.16	0.90
Threshold Adjusted ($\tau = 0.7$)	0.948	0.38	0.68
Strengthened RF (600 trees)	0.944	0.67	0.48

The baseline model exhibited high recall but extremely low precision, indicating substantial false positives. This would classify large urban areas as deficit zones, inflating policy response requirements.

The strengthened model improved class separability by:

- Increasing tree depth (capturing nonlinear spatial interactions)
- Increasing number of estimators (reducing variance)
- Applying threshold calibration

This improved the interpretive reliability of hotspot classification.

5.4.4 Why the Model Performed Well

Feature importance revealed that commercial POI density and accessibility centrality were dominant predictors. This aligns with accessibility theory (Hansen, 1959), which positions opportunity concentration as a primary mobility attractor.

Furthermore, spatial dependence theory (Anselin, 1988) explains why clustered demand surfaces generate predictable patterns. The model effectively learned stable morphological signals embedded in Dubai's urban structure rather than transient behavioral noise.

Thus, strong performance reflects structural regularity in land-use and mobility concentration rather than algorithmic overfitting.

5.4.5 Extending the Framework: YOLO and Sentinel-Based Detection

The literature review referenced deep learning object-detection architectures such as YOLO (You Only Look Once), originally introduced by Redmon et al. (2016), which enables real-time object detection within a single convolutional neural network framework.

YOLO-based architectures have since been widely adopted for vehicle detection in aerial and satellite imagery contexts due to their computational efficiency and strong localization performance.

While the present study utilized structured parking capacity data and POI proxies, an expanded framework could integrate high-resolution Sentinel-2 imagery or drone-derived orthophotos processed via YOLO-based object detection to estimate:

- Real-time vehicle occupancy,
- Surface-level parking utilization,
- Informal curbside saturation.

In such a hybrid model:

$$\text{Observed Occupancy}_i = \text{YOLO-detected Vehicles}_i$$

This extension would transform the current structural imbalance model into a dynamic occupancy monitoring system. Rather than relying solely on registered capacity values, the framework would incorporate pixel-level object recognition, providing empirical ground-truth validation of predicted hotspot zones.

Compared to purely statistical clustering methods, YOLO-enhanced systems enable fine-grained detection of parked vehicles and temporal variation. However, such integration introduces increased computational requirements, sensor dependency, and image-resolution constraints not required within the current scalable grid-based framework.

5.4.6 Toward Intelligent Governance: Linking ML to RAG

The following discussion is conceptual and is intended to position the empirical hotspot model within a broader decision-support architecture for urban parking governance.The predictive model identifies statistically significant deficit regimes. However, interpretation of these outputs requires contextual policy reasoning.

A Retrieval-Augmented Generation (RAG) layer could sit atop the classifier:

1. The model predicts hotspot probability.
2. Structured outputs are embedded as query vectors.
3. Relevant RTA policy documents are retrieved.
4. A language model generates calibrated planning recommendations.

Unlike traditional dashboards, RAG systems would transform spatial statistics into interpretable regulatory narratives.

This hybrid ML–RAG architecture represents a progression from predictive analytics to intelligent urban governance, where algorithmic detection is directly linked to policy knowledge bases.

5.4.7 AI Governance Stack and Decision Intelligence

Within the context of this thesis, the AI governance stack is best understood as a layered parking decision-support system in which the empirically implemented GIS and machine learning components form the analytical foundation for higher-level policy interpretation and governance reasoning.The integration of spatial statistics, machine learning, deep learning extensions, and RAG-based policy reasoning forms what may be conceptualized as an AI governance stack. This stack consists of five layers:

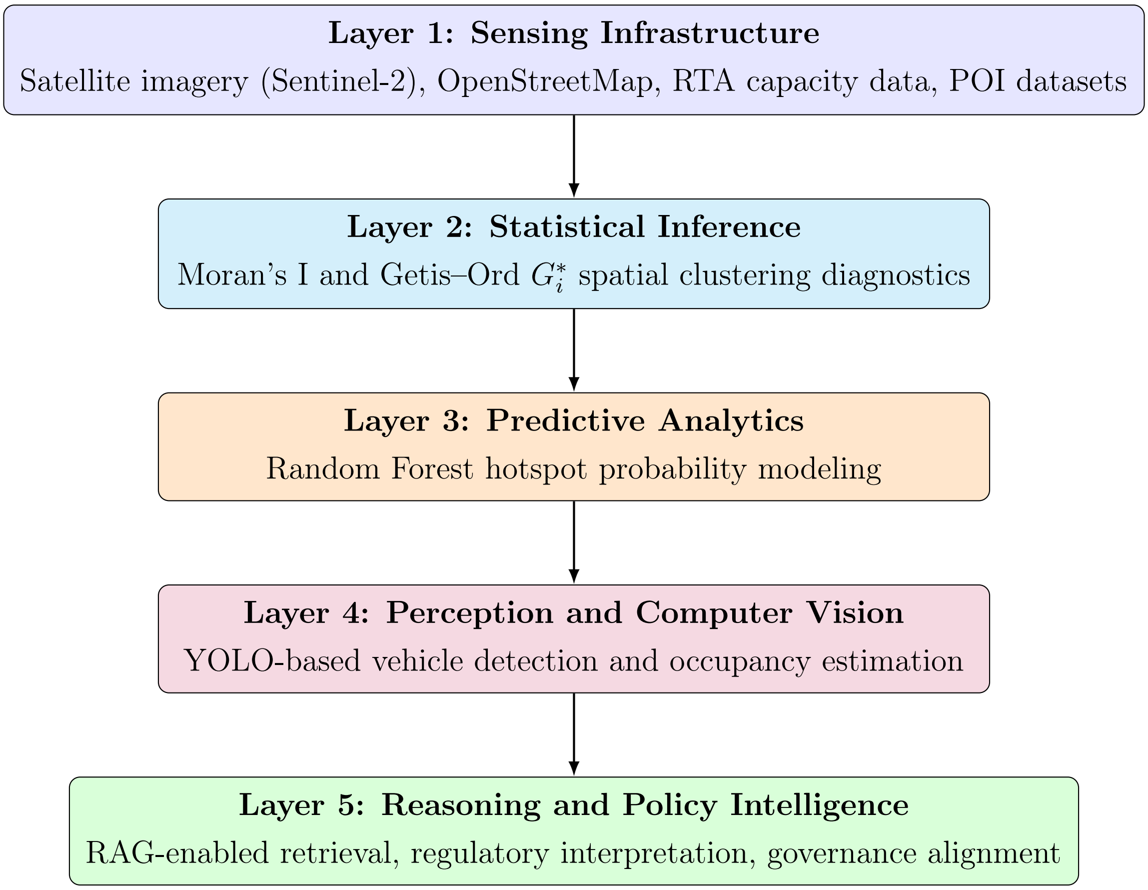


Figure 5.1: Multi-layer Urban AI Governance Stack integrating sensing, statistical inference, predictive modeling, perception, and reasoning layers.

Unlike conventional dashboards, this layered architecture enables end-to-end urban intelligence: from detection to contextual explanation.

5.4.8 Multi-Layer Urban AI Governance Stack

Accordingly, the stack is not presented as separate from the study's contribution, but as the extended decision-support logic through which parking imbalance detection can inform planning, regulation, and policy-oriented action.The proposed analytical framework can be conceptualized as a layered urban AI governance stack, extending beyond isolated predictive modeling toward structured decision intelligence.

Layer 1: Sensing Infrastructure The sensing layer aggregates heterogeneous data sources, including Sentinel-2 satellite imagery, OpenStreetMap vector datasets, and administrative parking capacity records. This aligns with digital twin paradigms in urban systems, where real-world infrastructure is continuously mirrored within computational environments.

Layer 2: Statistical Inference At the inferential layer, spatial autocorrelation statistics (Moran's I and Getis–Ord G_i^*) provide hypothesis-driven validation of clustered imbalance regimes (Getis and Ord, 1992; Anselin, 1995). This layer ensures that predictive modeling is grounded in statistically verified spatial dependence rather than purely algorithmic pattern recognition.

Layer 3: Predictive Analytics The predictive layer employs ensemble machine learning (Random Forest) to classify hotspot regimes based on engineered accessibility and land-use features. This transforms inferential diagnostics into probabilistic forecasting tools capable of scenario testing.

Layer 4: Perception and Computer Vision Deep learning extensions such as YOLO (Redmon et al., 2016) represent the perception layer, enabling pixel-level vehicle detection within high-resolution imagery. This layer bridges structural imbalance modeling with dynamic occupancy validation, supporting real-time calibration of inferred hotspot zones.

Layer 5: Reasoning and Policy Intelligence The reasoning layer integrates Retrieval-Augmented Generation (RAG) architectures (Lewis et al., 2020), grounding generative language models in retrieved policy documentation (e.g., RTA strategies,zoning codes, transit-oriented development guidelines).

Unlike traditional dashboards, this layer synthesizes quantitative outputs into governance-aligned narratives, enabling structured policy interpretation rather than raw data presentation.

Comparative Governance Context

Layered AI architectures have parallels in emerging smart city governance frameworks across jurisdictions.

European Union AI governance guidelines emphasize transparency, traceability, and human-in-the-loop validation for high-impact urban systems. Similarly, Singapore's Smart Nation initiative and Australia's Intelligent Transport Systems frameworks deploy multi-layer architectures separating sensing, analytics, and regulatory oversight.

The present framework aligns with these paradigms by:

- Preserving inferential transparency,
- Separating prediction from reasoning,
- Enabling auditability through retrieval grounding,
- Maintaining human review as a governance safeguard.

Positioning Within Urban Digital Twin Evolution

While Dubai's AI Lab initiatives focus primarily on macro-scale traffic congestion prediction, the framework developed in this thesis introduces a micro-scale parking intelligence layer that can complement broader digital twin systems.

By integrating sensing, inference, prediction, perception, and reasoning, the architecture advances toward a unified urban decision-support stack rather than a standalone predictive tool.

5.4.9 Limitations of the Current ML Framework

Despite strong performance, the model:

- Predicts structural imbalance, not real-time occupancy.
- Does not explicitly encode spatial dependence within tree structure.
- Relies on proxy demand indicators rather than behavioral microdata.

Future expansion incorporating YOLO-based occupancy detection, graph neural networks for network dependence, and domain adaptation for cross-city transfer would significantly extend predictive robustness.

5.5 Cross-City Transferability Insights

Preliminary transfer testing to Seattle revealed limited structural generalization. The Dubai-trained classifier produced a highly compressed probability distribution, assigning uniformly low hotspot probabilities across the Seattle grid.

This suggests that learned spatial relationships are context-dependent and sensitive to local urban morphology, transport hierarchy, and land-use intensity patterns.

Rather than indicating model failure, this outcome highlights spatial non-stationarity in urban parking dynamics and reinforces the importance of localized calibration

5.5.1 Spatial Non-Stationarity and Domain Sensitivity

The cross-context application of the Dubai-trained classifier to Seattle produced a compressed probability distribution and near-uniform low hotspot likelihoods across the Seattle grid. This outcome is consistent with *spatial non-stationarity*: the relationship between accessibility, land-use intensity, and parking supply is not invariant across cities, but depends on local urban morphology, regulatory conditions, and behavioural parking dynamics.

Morphological and network differences (Dubai vs. Seattle). Dubai exhibits a highly centralized coastal economic corridor with strong commercial agglomeration and major arterial accessibility, producing concentrated demand attractors and structurally clustered imbalance surfaces. In contrast, the Seattle application area reflects a more polycentric and neighbourhood-scaled urban form with different road hierarchy, block structure, and parking typologies (e.g., distributed curb supply and mixed off-street provision). Because the model learns reduced-form associations between engineered features (e.g., POI densities, road density, and accessibility distances) and hotspot labels, a shift in the joint distribution of these features can reduce predictive separability under transfer.

Zoning and land-use regulation regimes. Planning and zoning frameworks influence how activity density, curb allocation, and off-street requirements co-evolve. In Dubai, rapid mixed-use development and intensive corridor investment can yield sharp spatial gradients in opportunity concentration. In many North American contexts, zoning constraints, residential permit systems, and minimum/maximum parking requirements can reshape the spatial mapping between activity proxies and observed supply. These policy differences contribute to domain shift, meaning that even if the same features are computed at the same grid resolution, their behavioural interpretation differs across cities.

Parking regulation and pricing structures. Parking markets generate spillovers through search, substitution, and displacement: scarcity in one cell can shift search into adjacent cells, amplifying local pressure. Such processes are sensitive to enforcement intensity, time limits, pricing structures, and availability of park-and-ride alternatives. Parking search models and cruising externalities imply that regulation can materially alter the spatial signature that the hotspot statistic detects (Tang et al., 2014; Shoup, 2006). Consequently, a classifier trained on Dubai's regime may not transfer without recalibration when enforcement and pricing incentives differ.

Car dependency and modal substitution. The strength of the accessibility–parking relationship depends on mode shares and the feasibility of substitution to rail or bus for activity corridors. Where transit coverage, pricing, or first/last-mile connectivity differs, POI-based attraction fields may translate into different parking search pressure and capacity patterns. This affects both the imbalance surface and the inferential Gi* labels used as the learning target.

Implications for modeling and evaluation. The transfer experiment therefore supports two methodological conclusions. First, cross-context deployment should be treated as a *distribution-shift* problem rather than a simple out-of-sample prediction task, motivating local calibration, feature normalization, and potentially additional morphology-based covariates.

Second, because hotspot labels are inferential outputs contingent on spatial weights and scale, sensitivity checks to grid resolution and weights remain important for robust interpretation (Openshaw, 1984; Anselin, 1988).

5.6 Theoretical Implications

The findings reinforce spatial equilibrium theory and central place logic: parking imbalance emerges where accessibility gradients intersect with economic agglomeration.

More broadly, the results suggest that urban parking behavior and decision-making can be interpreted through observed spatial structures, where concentrated opportunity fields, constrained supply, and network accessibility jointly shape localized parking pressure.

The study further demonstrates that grid-based spatial standardization reduces MAUP bias while enabling machine learning integration.

5.7 Methodological Implications

The integration of:

- Grid-based tessellation
- Spatial autocorrelation statistics
- Ensemble classification
- Morphological satellite validation
- Conceptual RAG interpretability layer

represents a unified GIS–AI analytical framework.

This approach bridges descriptive spatial statistics and predictive modeling within a reproducible PyQGIS environment, while also supporting the interpretation of parking-related decision environments through a unified spatial analytical framework.

Chapter 6

Conclusion

This study examined urban parking behavior and decision-making in Dubai through a grid-based analytical framework integrating GIS, spatial statistics, and supervised machine learning.

By combining spatial autocorrelation analysis, engineered accessibility features, and ensemble classification methods, the research demonstrates that:

- parking imbalance follows statistically significant spatial clustering patterns,
- demand proxies derived from POI density provide meaningful insight into structural parking pressure,
- accessibility and network centrality are key determinants of parking imbalance formation, and
- supervised models can identify hotspot zones using structured spatial features.

These findings show that urban parking behavior and decision-making may be meaningfully interpreted through observed spatial patterns of supply, accessibility, and activity intensity. In this thesis, these components are brought together within a single analytical and decision-support perspective, allowing parking imbalance to be examined not merely as a technical allocation issue, but as part of a broader urban governance problem.

The methodological integration of PyQGIS preprocessing, spatial statistics (Moran's I and Getis–Ord G_i^*), and Random Forest classification provides a reproducible analytical framework for urban parking assessment and decision support.

Importantly, the cross-city transfer experiment suggests that spatial determinants of parking imbalance are city-specific and require contextual calibration for generalization.

This work contributes to the intersection of geospatial analytics, machine learning, urban mobility planning, and AI-enabled decision support by showing how parking imbalance diagnostics can inform policy interpretation and governance-oriented action.

This work contributes to the emerging intersection of geospatial analytics, machine learning, and urban mobility planning.

6.1 Policy and Planning Implications

Based on the observed core–periphery hotspot structure in Dubai, the engineered feature diagnostics, and the precision-oriented strengthened classifier, the following recommendations are proposed. Each recommendation is independent and can be implemented incrementally.

- **Hotspot-targeted pricing and time-limit refinement (core corridors).** Adopt zone-based pricing and time-limit policies that explicitly prioritize statistically significant hotspot corridors (high G_i^*). This reduces cruising, improves turnover, and targets the externalities of search traffic documented in parking economics and search models (Shoup, 2006; Tang et al., 2014). In operational terms, the model outputs can be used to rank hotspot cells for pilot implementation and monitoring.
- **Shift long-stay parking away from the CBD through peripheral supply and park-and-ride integration.** Where oversupply or low-demand zones are detected, designate peripheral facilities for longer-stay parking and integrate them with rail and bus access to reduce CBD-bound car trips. This recommendation is consistent with the thesis results showing concentrated deficit in the northern corridor and oversupply patterns elsewhere, and it aligns with a demand-management strategy rather than expanding supply in already constrained cores.
- **Adopt sensor-enabled curb management and reservation-based parking tools as a best-practice benchmark for future upgrades.** Although Dubai provides structured capacity inventories, many advanced cities complement inventories with real-time occupancy sensing, digital payments, and reservation functionality to reduce uncertainty and cruising. A phased approach can

begin with pilot corridors inside the identified hotspot band, using occupancy sensing to validate model outputs and improve the responsiveness of pricing and enforcement. This directly strengthens the validation layer of the framework beyond qualitative imagery checks.

Land-use coordination in persistent hotspots (shared parking and mixed-use bundling). In dense mixed-use clusters where commercial and institutional POI densities are high, promote shared parking agreements and coordinated loading/short-stay design to reduce peak conflicts. This is supported by the feature-importance findings showing dominance of activity intensity and accessibility factors in hotspot classification.

- **Use the trained classifier as an operational screening tool for RTA deployment decisions.**

  A practical pathway for adoption is to embed the strengthened model into a GIS dashboard:

  (i) ingest updated capacity/POI/network layers,

  (ii) recompute features

  (iii) predict hotspot likelihood, and

  (iv) generate a ranked list of priority cells for field inspection, enforcement targeting, or infrastructure intervention. This converts the model from an academic predictor into a repeatable decision-support workflow.

- **Transferability protocol for external cities (domain calibration)**

  For use outside Dubai, implement a minimum transfer protocol:

  (i) Recompute the grid in the appropriate local UTM projection to ensure metric consistency.

  (ii) Normalize predictors using local distributional properties rather than parameters derived from the Dubai dataset.

  (iii) Recompute Getis–Ord G_i^* hotspot labels within the local spatial context prior to supervised learning.

  (iv) Retrain or fine-tune the classifier using a small locally labeled subset to recalibrate decision boundaries.

  This recommendation is justified by the observed domain sensitivity in the Seattle transfer results and by the dependence of spatial inference on MAUP and spatial weight specification dependence in spatial inference (Openshaw, 1984; Anselin, 1988).

6.2 Operational Deployment Framework for RTA Integration

While this study develops a spatial-statistical and machine learning framework for detecting structural parking imbalance, its practical value lies in its potential institutional deployment. The Roads and Transport Authority (RTA) could adapt the developed model into a decision-support module integrated within its smart mobility infrastructure.

6.2.1 Grid-Based Parking Intelligence Layer

The 250 m grid framework developed in this thesis provides a standardized spatial unit compatible with GIS-based municipal dashboards. RTA already operates digital mapping platforms for zone management and enforcement. By overlaying the imbalance index and Gi* hotspot classifications onto existing zoning layers, planners could obtain a live spatial intelligence layer identifying statistically significant deficit and oversupply regimes.

This grid-based architecture mitigates administrative boundary bias and aligns with spatial econometric principles of standardized aggregation (Anselin, 1988; Openshaw, 1984).

6.2.2 Data Pipeline Integration

The model can be operationalized through a structured data pipeline:

1. Periodic ingestion of updated parking capacity records.
2. Automated extraction of POI and land-use updates from open geospatial sources.
3. Recalculation of imbalance index and spatial autocorrelation statistics.
4. Execution of the trained Random Forest classifier.

The Random Forest model requires only engineered spatial features and does not depend on proprietary sensor infrastructure. This makes it scalable and cost-efficient.

6.2.3 Dynamic Tariff Adjustment Mechanism

Using predicted hotspot probabilities, RTA could implement threshold-based tariff triggers. For example:

$$P(\text{Hotspot}) > 0.75 \Rightarrow \text{Premium Dynamic Pricing Tier}$$

Such probability-based pricing tiers would ensure that tariff escalation is grounded in statistically validated structural imbalance rather than subjective zoning designation. This aligns with economic models of congestion pricing and cruising mitigation (Shoup, 2006).

6.2.4 Strategic Zoning Feedback Loop

Model outputs can inform planning approval processes. Before authorizing new commercial developments, planners could evaluate projected imbalance shifts by simulating additional POI density within affected grid cells. If predicted hotspot probability exceeds a defined threshold, parking provision ratios or shared parking mechanisms could be recalibrated.

This transforms the model from reactive congestion analysis to proactive planning control.

6.2.5 Transit Coordination Module

Because accessibility and network centrality were dominant predictive features, the model can support transit-oriented development policy. RTA may use hotspot outputs to identify corridors where parking supply should be constrained in favor of transit usage, reinforcing modal shift objectives.

6.2.6 Continuous Monitoring Dashboard

A GIS dashboard integrating:

- Imbalance Index heatmap
- Gi* statistical clusters

- Random Forest hotspot probabilities
- Tariff zone overlays

would enable monthly or quarterly review cycles. Model retraining can be scheduled annually to capture evolving land-use intensity patterns.

6.2.7 Institutional Value Proposition

The proposed framework offers RTA:

- Evidence-based tariff calibration
- Reduced reliance on expensive sensor deployment
- Proactive zoning risk assessment
- Cross-sector integration between parking, land-use, and transit planning

Importantly, the cross-city transfer experiment demonstrates that model calibration must remain context-sensitive. Therefore, institutional deployment should include periodic retraining using Dubai-specific data to preserve predictive validity. This study contributes methodologically by:

- Implementing a 250 m uniform tessellation to mitigate MAUP bias.
- Combining spatial autocorrelation statistics with supervised classification.
- Introducing structured demand proxies via POI categorization.
- Demonstrating the limitations of cross-city transfer without recalibration.
- Integrating PyQGIS automation for reproducible spatial modeling.

6.3 Future Research Directions

The present study establishes a scalable grid-based framework integrating spatial statistics and supervised machine learning for structural parking imbalance detection. However, multiple avenues exist for methodological expansion and system integration.

6.3.1 Integration of Real-Time Object Detection (YOLO + Sentinel)

Future work may incorporate deep learning object detection architectures,such as YOLO (Redmon et al., 2016), applied to high-resolution Sentinel-2 imagery or drone-derived orthophotos. Such integration would enable:

- Direct vehicle counting at grid-cell level,
- Temporal occupancy estimation,
- Validation of statistically inferred hotspot zones.

By replacing static capacity values with dynamically observed occupancy:

$$\text{Dynamic Imbalance}_i = Z(\text{Detected Vehicles}_i) - Z(\text{Capacity}_i),$$

the model could transition from structural imbalance assessment toward real-time congestion intelligence.

6.3.2 Graph Neural Networks for Explicit Spatial Dependence

While Random Forest classifiers capture nonlinear relationships, they do not explicitly encode network adjacency within their learning structure. Future research could implement Graph Neural Networks (GNNs) to directly model grid-cell contiguity and road-network topology.

Such architectures would allow the learning function to incorporate spatial weight matrices intrinsically, rather than relying on separate Moran's I diagnostics.

6.3.3 Domain Adaptation and Multi-City Joint Training

The Seattle transfer experiment demonstrated spatial non-stationarity. Future research may employ domain adaptation frameworks to recalibrate models across heterogeneous urban morphologies.

Adversarial transfer learning or multi-city joint training could enhance generalizability while preserving localized calibration.

6.3.4 Retrieval-Augmented Governance Systems

A significant extension involves embedding predictive outputs within a Retrieval-Augmented Generation (RAG) framework.

Rather than producing static hotspot maps, the system could:

1. Retrieve policy documents, zoning codes, and RTA regulations,
2. Contextualize predicted imbalance zones,
3. Generate structured planning recommendations.

This would transform the analytical framework into a decision-support system integrating spatial detection, machine reasoning, and regulatory knowledge retrieval.

6.3.5 Toward an Integrated Urban AI Stack

Ultimately, future research may unify:

- Spatial statistics (Moran's I, Gi*),
- Supervised machine learning (Random Forest),
- Deep learning object detection (YOLO),
- Knowledge retrieval systems (RAG),
- Policy reasoning layers.

Such an integrated architecture would represent a full-stack urban AI governance system capable of detecting, interpreting, and responding to parking imbalance dynamically.

Recent announcements regarding Dubai's AI-driven congestion prediction initiatives indicate strong institutional commitment to intelligent transport systems. However, existing initiatives focus primarily on macro-scale traffic flow modeling.

The present study complements such efforts by introducing a micro-scale,grid-based parking imbalance detection framework, thereby extending intelligent mobility governance into the parking domain.

Overall, the thesis demonstrates that urban parking behavior and decision-making can be examined through the conjoint use of spatial inference, predictive modeling, and decision-support interpretation. In this respect, the study offers not only an empirical assessment of parking imbalance in Dubai, but also a foundation for a broader urban parking decision-support system capable of linking spatial evidence to governance and policy response.

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Appendices

Appendix A

Python Implementation (ME5)

This appendix contains the full Python implementation used for feature engineering, spatial statistics, and supervised modeling. The full executable notebook is available in the digital submission package under: $/Supplementary/ME5_Notebook.ipynb$

A.1 Data Preprocessing

A.2 Spatial Autocorrelation Computation

A.3 Random Forest Training

Appendix B

Supplementary Figures

B.1 Additional Correlation Diagnostics

B.2 Extended Feature Importance Plots

Appendix C

GIS Project Files

C.1 QGIS GeoPackage Layers

The GeoPackage (.gpkg) used in this study contains:

- 250 m spatial grid
- Engineered feature layers
- Hotspot classification layer
- Transferability outputs

C.2 KML and Raw Spatial Data

Original RTA KML parking datasets and OpenStreetMap extracts are archived separately in the digital submission folder.

Appendix D

AI Usage Declaration

This study utilized AI-assisted tools for language refinement, code optimization, and conceptual structuring. All analytical results, modeling decisions, and interpretations remain the author's original work.